



USER INTERFACE DESIGN

23CSE113

LAB MANUAL

SUBMITTED BY	
NAME	Padmanabhuni Akhil
ROLL NO	AV.SC.U4CSE25223
YEAR/SEM/SECTION	1ST YEAR/ SEM-2/CSE-B
CREDITS	03

SUBMITTED TO	DR. RAJ KUMAR BATCHU
DESIGNATION	ASST.PROFFESSOR (Sl. Grade)
DEPARTMENT	SCHOOL OF COMPUTING

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Week-3	3. A. Create the class time table using HTML Elements. B. Create an HTML Webpage displaying Syllabus of all the subjects with the timetable.	16-29	
Week-4	4.A. A college wants to collect student details through an online admission form. Design a registration form using only HTML tags (no CSS) with proper labels, table alignment, background color, and borders. B. A company wants to design a webpage that displays two different forms side by side using iframes. One iframe should display a Login Form and the other should display a Registration Form.	30-36	
Week-5	A. Creating a card layout using <div> tag consider three cards and a button using html and CSS B. Navigation using iframe (mini website) upon clicking each buttons the details about that button should be display on the same page C. Design amrita login page by showing different campus on home page upon clicking on each campus the details about each campus should	36-53	

	be display on the same page use html and css to design a stylish webpage.		
	6.A. Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. B. Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. Based on the category we select corresponding products should be displayed on the sidebar.	54-69	
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Week-9	9.Create a webpage using HTML, CSS, JS based on the give conditions: a) Create a button for each program, based on the button, clicked the corresponding program has to be executed and should display the output. b) The logics for each button includes finding an even or odd number, prime or not, factorial of a number, Armstrong or not. c)Use appropriate design to display the output.	79-82	
Week-10	10.A. Write JavaScript conditional statement to find the sign of product of three numbers using functions. Display an alert box with specified sign. B. Write a java script conditional statements to sort 3 numbers using functions display an alert box to show the result. C. Write a java script conditional statement to find largest of find largest of five numbers using function	83-87	

	and onclick event display an alert box to show the result. D. Write a java script for loop that will iterate from 0 to 15. For each iteration, it will check if the current number odd or even, and display a message on screen. E. Write a java script program to read 6 subjects marks from the student then computes the average marks of students then, this average is used to determine the corresponding grade.		
Week-11	11. A. A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters. B. A registration form should validate email format before submission. If invalid, the form should not be submitted. C. A system should check password strength while typing and display whether it is weak or strong. D. A form should validate a mobile number when the user leaves the input field. The number must be exactly 10 digits. E. Design an HTML page, such that the registration form must validate: Name (not empty) Email (valid format) Password (minimum 6 characters)	88-96	
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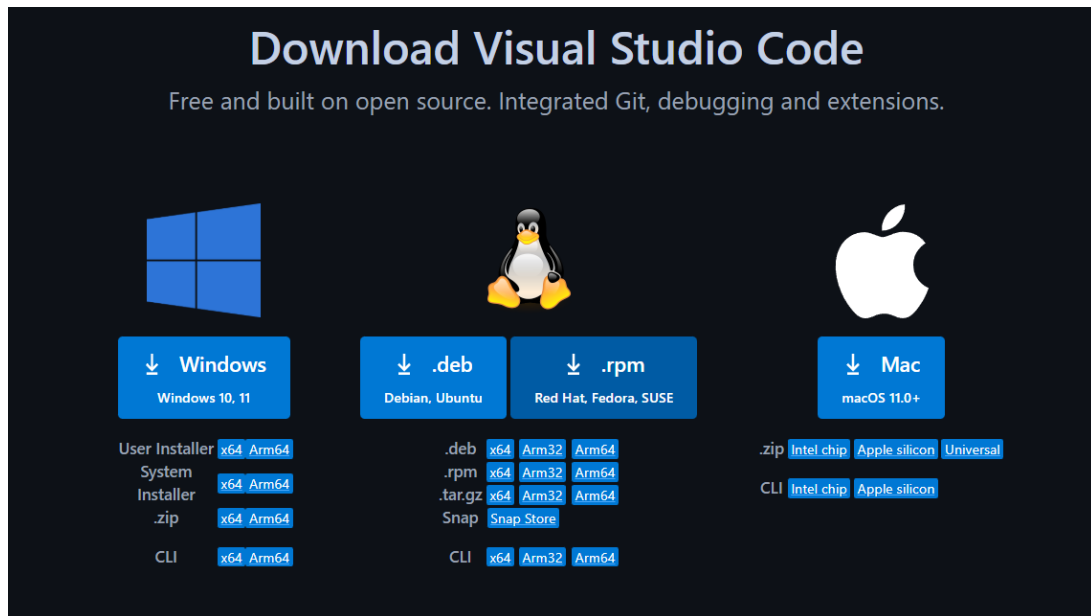
WEEK -1

1. A. AIM: Steps to install VS CODE

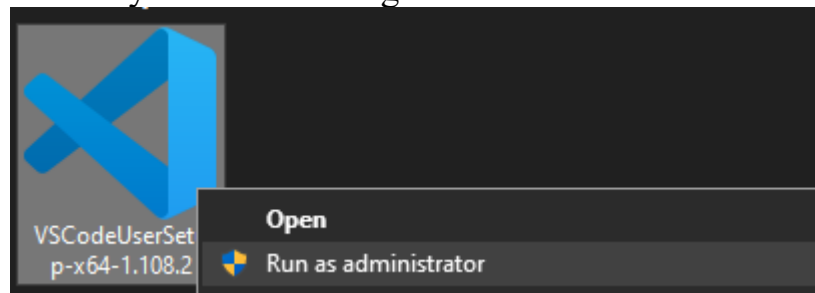
PROCEDURE:

STEP-1:- GO TO THE WEBSITE -<https://code.visualstudio.com/Download>

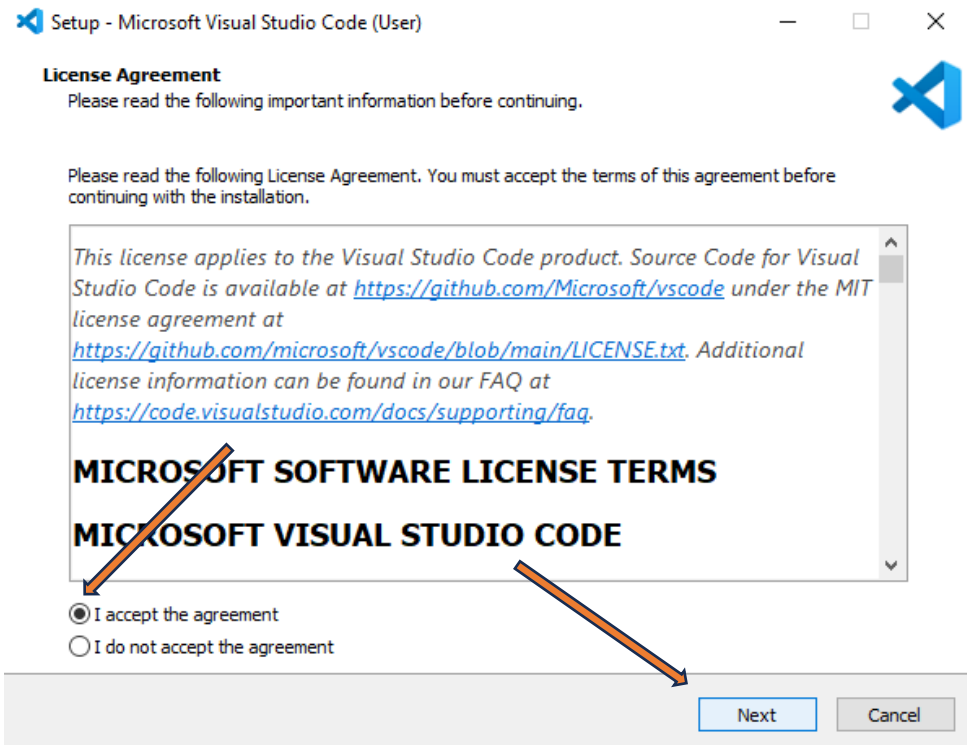
STEP-2: Download The Windows Version



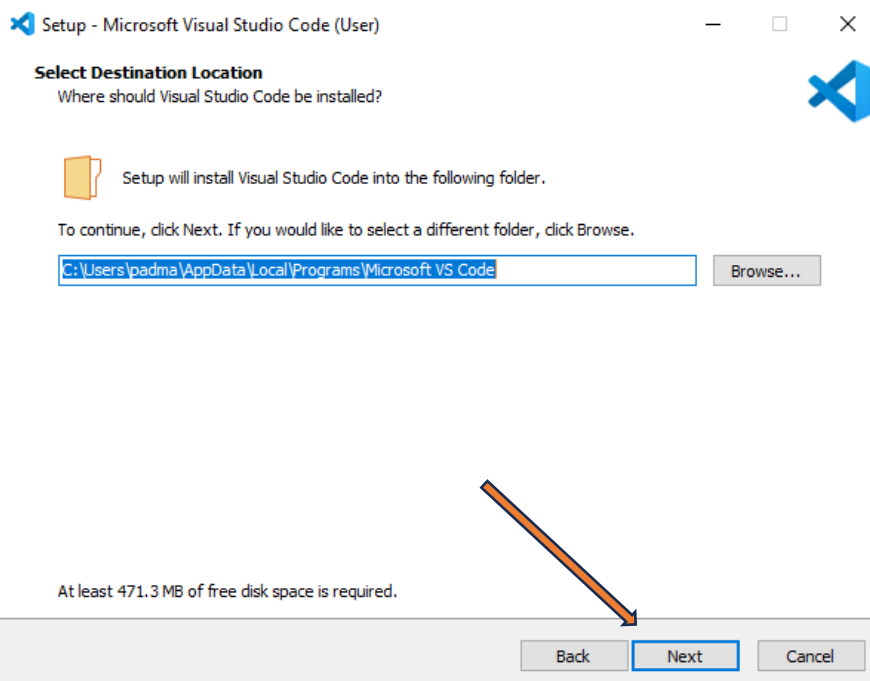
STEP-3: Run the file by Double-clicking on it



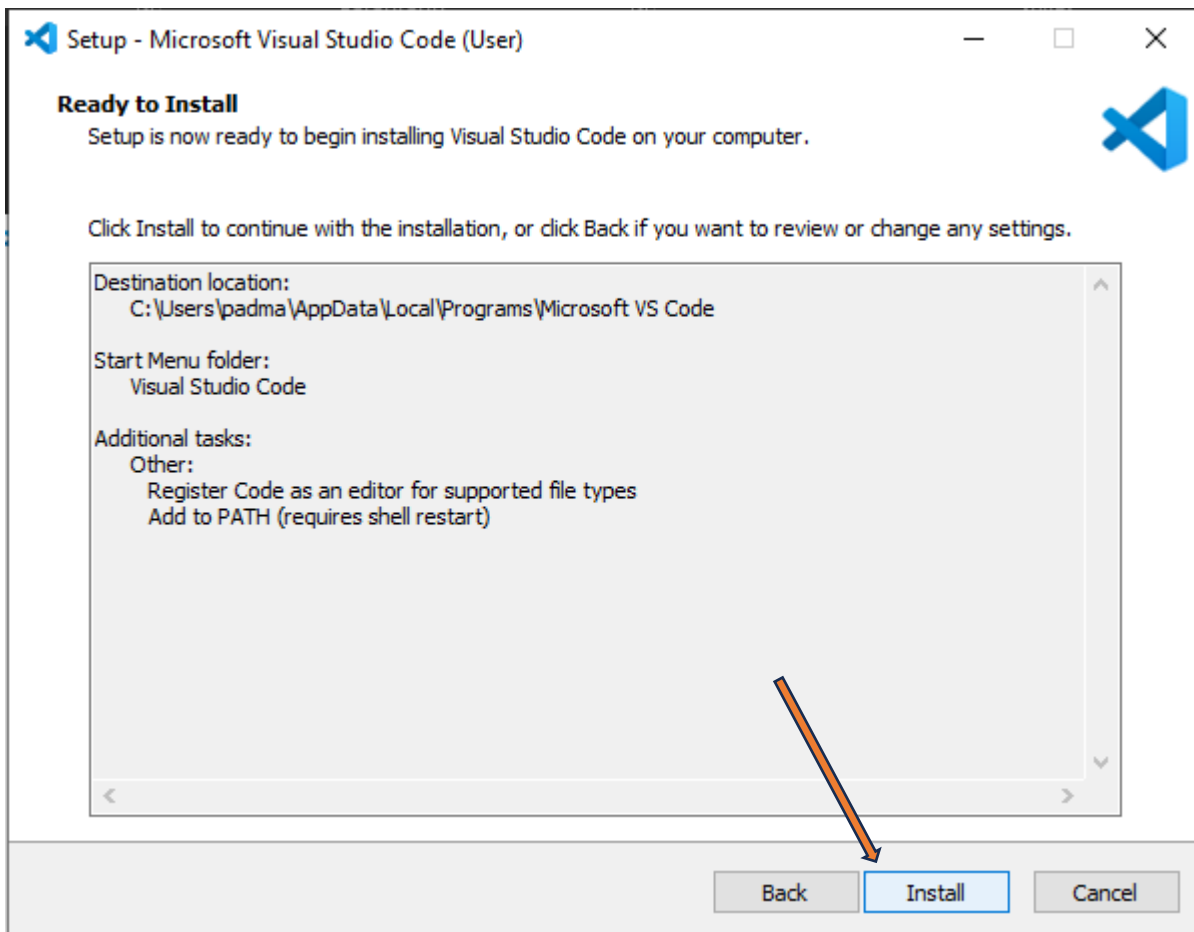
STEP-4



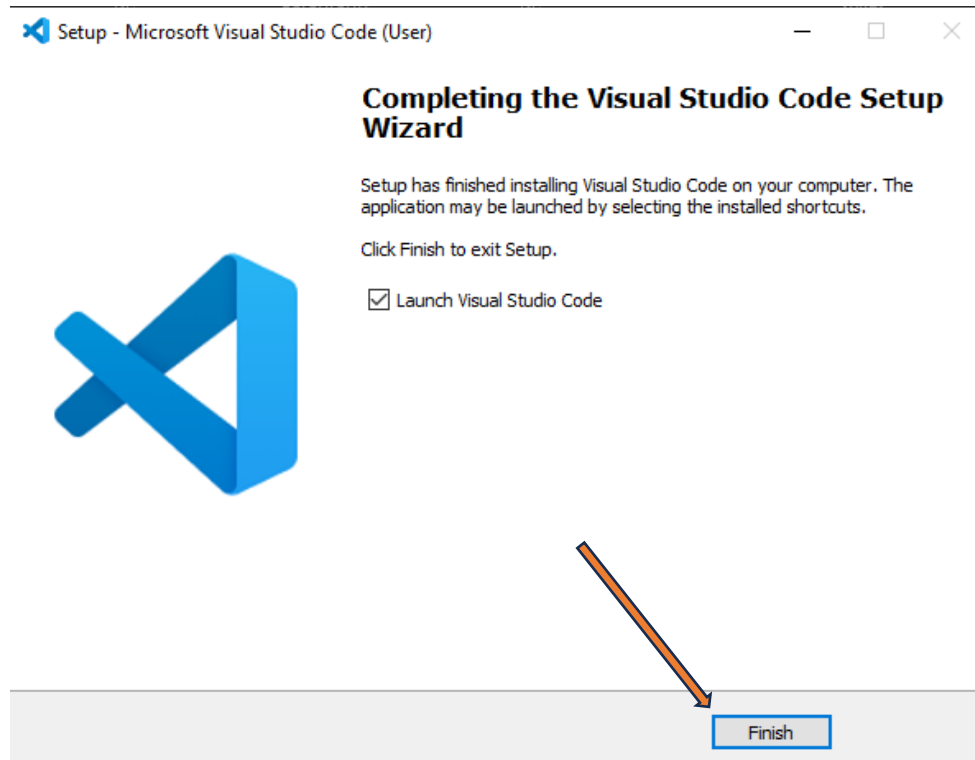
STEP-5:



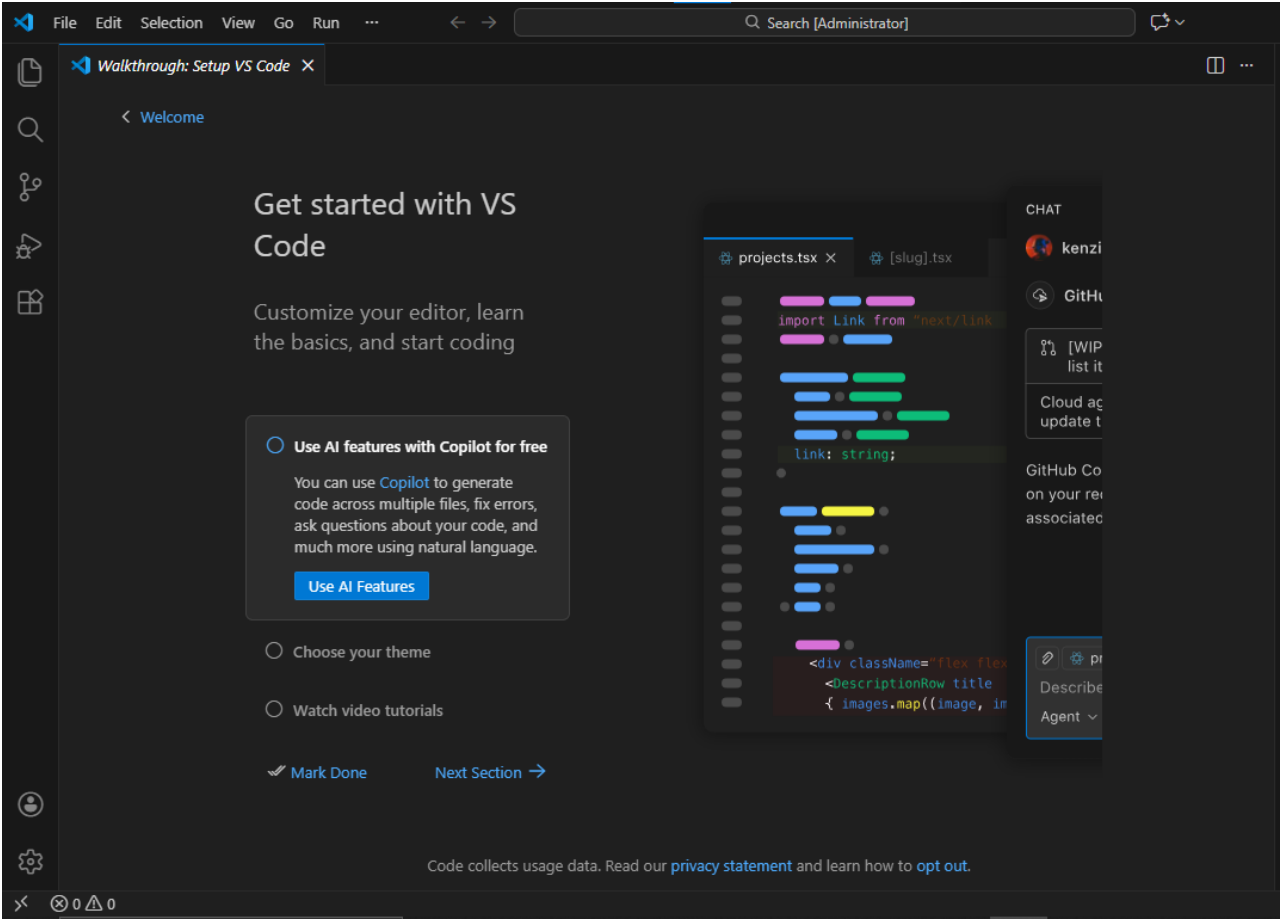
STEP-6:



Step-7:



Now, It is Ready to use



B. AIM: Create a Basic HTML Page

Program:

```
UID-AV.SC.U4CSE25223
<!DOCTYPE html>
<html>
  <head>
    <title>First Web Page</title>
  </head>
  <body>
    <h1>Happy to learn HTML</h1>
    <p>This is my first sample code</p>
  </body>
</html>
```

OUTPUT:

UID-AV.SC.U4CSE25223

Happy to learn HTML

This is my first sample code

Explanation of the tags:

- HTML stands for Hyper Text Markup Language.
- It tells the web browser what to display and how to display it.
- <DOCTYPE HTML>: The <DOCTYPE html>element tells the browser, this is an HTML5 document.
- <html>: The <html> element is the root element of an HTML page.
- <head>: The <head> element contains meta information about the HTML page.
- <title>: The <title> element specifies a title for the HTML page.
- <body>: The <body>element contains all visible content like text, images, links, tables, etc.
- <h3>: The <h3> element defines a large heading of size in the scale 1 to 6.
- <p>: The <p> element defines a paragraph.

C. AIM: Basic HTML Tags

Program:

-
 and style attribute in tags

```
<!DOCTYPE html>
<html>
  <head>
    <title>Student Details</title>
  </head>
  <body>
    <h1 style="color:red">My Details</h1>
    <p style="color:aquamarine">Name: Padmanabhuni Akhil <br> Roll No. AV.SC.U4CSE25223
<br>Section: CSE-C<br>University: Amrita</p>
    <h2 style="color:greenyellow">Personal Details</h2>
    <p style="color:brown">Name: Padmanabhuni Akhil <br> Roll No. AV.SC.U4CSE25223
<br>Section: CSE-C<br>University: Amrita</p>
  </body>
</html>
```

Output:

My Details

Name: Padmanabhuni Akhil
Roll No. AV.SC.U4CSE25223
Section: CSE-C
University: Amrita

Personal Details

Name: Padmanabhuni Akhil
Roll No. AV.SC.U4CSE25223
Section: CSE-C
University: Amrita

- URL Tags Text
- Img Tags

UID-AV.SC.U4CSE25223

```
<!DOCTYPE html>
<html>
  <head>
    <title>Links</title>
  </head>
  <body>
    <h2>HTML Links</h2>
    <p>HTML Links for some websites</p>
    <h2>Amrita Vishwa Vidyapeetham</h2>
    <p>Deemed to be a University</p>
    
    <a href="https://www.amrita.edu/">Link</a>
    <h2>SRM</h2>
    <p>Deemed to be a University</p>
    
    <a href="https://www.srmist.edu.in/">Link</a>
    <h2>VIT</h2>
    <p>Deemed to be a University</p>
    
    <a href="https://vit.ac.in/">Link</a>
  </body>
</html>
```

Output:

UID-AV.SC.U4CSE25223

HTML Links

HTML Links for some websites

Amrita Vishwa Vidyapeetham

Deemed to be a University



SRM

Deemed to be a University



VIT

Deemed to be a University



Explanation of the tags:

- `<br>`: The HTML `<br>` element defines a line break. it is an empty tag with no end tag.
- `Style`: The HTML `style` attribute is used to add styles to an element such as color, font, size and more.
- `<img>`: The `<img>` tag is a fundamental HTML element used to embed an image into a web page.
- `<a>`: The HTML `<a>` tag ,which stands for “anchor”, is used to define a hyperlink.

WEEK-2

2.

Code:

```
<!DOCTYPE html>
<html>
UID-CSE25223
<head>
  <title>Life Quote</title>
</head>
<body background="Webpage.jpg" style="background-size: cover;">
  <table width="98%" border="4" style="text-align: center;border-color: blueviolet;font-size: larger;font-family: 'Times New Roman', Times, serif;color: brown;">
    <tr>
      <th style="border-color: transparent;">Life Quotes</th>
    </tr>
  </table>
  <pre style="text-align: center;color: blueviolet;font-size:large;">      You are the
sum total of everything you've ever seen,

  heard, eaten, smelled, been told, forgot-it's all there.

  Everything influences each of us, and because of that I try

  to make sure that my experiences are positive." -Maya Angelou</pre>
  <table width="800" border="1" style="text-align: center;font-family: 'Times New Roman',
Times, serif;background-image: url(Amaravati.jpg);background-repeat:no-repeat;background-size: cover;" align="center" cellspacing="10" cellpadding="10">
    <caption style="background-color: azure;font-size: x-large;font-family: Arial, Helvetica, sans-serif;">AMRITA VISHWA VIDYAPEETHAM</caption>
    <tr style="background-color: aqua;">
      <th>S.No</th>
      <th>Campus</th>
      <th>Location</th>
      <th>Contact No</th>
      <th colspan="2">Contact Person</th>
    </tr>
    <tr>
      <th style="background-color: aquamarine;">1</th>
      <th style="background-color: aquamarine;">Amaravati</th>
      <th style="background-color: azure;">Kuragallu</th>
      <th rowspan="2">+91-9807654321</th>
      <th>Rajkumar</th>
      <th>Batchu</th>
    </tr>
    <tr>
      <th style="background-color: aquamarine;">2</th>
```

```

<th style="background-color: aquamarine;">Amritapuri</th>
<th style="background-color: azure;">Amrita Nagar</th>
<th colspan="2">Adithya Ram</th>
</tr>
<tr>
<th style="background-color: aquamarine;">3</th>
<th style="background-color: aquamarine;">Coimbatoore</th>
<th style="background-color: azure;">Ettimadai</th>
<th >+91-9087654321</th>
<th>AAshish</th>
<th style="background-color: rgb(129, 26, 220);">B</th>
</tr>
</table>
</body>
</html>

```

Output:

UID-CSE25223

Life Quotes

You are the sum total of everything you've ever seen,
heard, eaten, smelled, been told, forgot--it's all there.
Everything influences each of us, and because of that I try
to make sure that my experiences are positive." -Maya Angelou

AMRITA VISHWA VIDYAPEETHAM

S.No	Campus	Location	Contact No	Contact Person	
1	Amaravati	Kuragallu	+91 9807654321	Rajkumar	Batchu
2	Amritapuri	Amrita Nagar		Adithya Ram	
3	Coimbatoore	Ettimadai	+91 9087654321	AAshish	B

Explanation of the tags:

- `<pre>`: Shows text exactly as you type it, keeping all spaces and line breaks.
- `<caption>`: Adds a short title above a table.
- `rowspan`: Makes one table cell stretch down across several rows.
- `colspan`: Makes one table cell stretch across several columns.
- `background`: Sets a background image for an element (old way; now we usually use CSS).

WEEK-3

3.A. Aim: To create the class Timetable using HTML Elements.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>II CES-C Mater TT</title>
</head>
<body>
  <table border="3" style="font-family: 'Courier New', Courier, monospace;text-align:
center;" cellspacing="0" cellpadding="10">
    <tr style="background-color: yellow;">
      <th colspan="12">UID-CSE25223 Master Time Table</th>
    </tr>
    <tr>
      <th rowspan="2">Time/ Day</th>
      <th>Slot 1</th>
      <th>Slot 2</th>
      <th style="background-color: yellow;" rowspan="2">10:35-10:45</th>
      <th>Slot 3</th>
      <th>Slot 4</th>
      <th>Slot 5</th>
      <th style="background-color: yellow;" rowspan="2">1:15-2:10</th>
      <th>Slot 6</th>
      <th style="background-color: yellow;" rowspan="2">3:00-3:10</th>
      <th>Slot 7</th>
      <th>Slot 8</th>
    </tr>
    <tr>
      <th>08:55-09:45</th>
      <th>09:45-10:35</th>
      <th>10:45-11:35</th>
      <th>11:35-12:25</th>
      <th>12:25-01:15</th>
      <th>02:10-09:45</th>
      <th>03:10-04:00</th>
      <th>04:00-04:50</th>
    </tr>
    <tr>
      <th>Monday</th>
      <td colspan="2" style="background-color: aqua;">23MAT116 Discrete
Mathematics_lab2</td>
      <th rowspan="5" style="background-color: yellow;vertical-align:bottom;writing-
mode:sideways-lr;">Short Break</th>
      <td>23PHY115 Modern Physics</td>
      <td>23CSE111 Object Oriented Programming</td>
```

```

        <td>22ADM111 Glimpses of Glorious India</td>
        <th rowspan="5" style="background-color: yellow;vertical-align:bottom;writing-
mode:sideways-lr;">Lunch Break</th>
        <td>22ADM111 Glimpses of Glorious India(P)</td>
        <th rowspan="5" style="background-color: yellow;vertical-align:bottom;writing-
mode:sideways-lr;">Short Break</th>
        <td colspan="2">Library/Sports</td>
    </tr>
    <tr>
        <th>Tuesday</th>
        <td colspan="2" style="background-color: rgb(227, 84, 235);">23CSE113 User
Interface Design LAB(BYOD)</td>
        <td>23MAT117 Linear Algebra</td>
        <td>23MAT116 Discrete Mathematics</td>
        <td style="color: red;">COUNSELLING</td>
        <td style="background-color: bisque;">23MEE115 Manufacturing Practice LAB</td>
        <td colspan="2" style="background-color: bisque;">23MEE115 Manufacturing
Practice LAB</td>
    </tr>
    <tr>
        <th>Wednesday</th>
        <td >23PHY115 Modern Physics</td>
        <td>23CSE113 User Interface Design</td>
        <td colspan="2" style="background-color: aquamarine;">23MAT117 Linear
Algebra_LAB(BYOD)</td>
        <td style="color: red;">COUNSELLING</td>
        <td >Library</td>
        <td>22ADM111vGlimpses of Glorious India</td>
        <td>Library/Sports</td>
    </tr>
    <tr>
        <th>Thursday</th>
        <td >23CSE111 Object Oriented Programming</td>
        <td>23MAT116 Discrete Mathematics</td>
        <td colspan="2" style="background-color: blueviolet;">23CSE111Object Oriented
Programming _Lab2</td>
        <td >23MAT117 Linear Algebra</td>
        <td >Library</td>
        <td colspan="2">Library/Sports</td>
    </tr>
    <tr>
        <th>Friday</th>
        <td >23CSE111 Object Oriented Programming</td>
        <td>23MAT116 Discrete Mathematics</td>
        <td >23PHY115 Modern Physics</td>
        <td >23PHY115 Modern Physics</td>
        <td >23CSE113 User Interface Design</td>
        <td >Library</td>

```

```

        <td colspan="2">Library/Sports</td>
    </tr>
</table>
<p><br><br><br><br></p>
<table border="3" style="font-family: 'Courier New', Courier, monospace;"
cellspacing="0" cellpadding="10">
    <tr>
        <th>Course Code</th>
        <th>Course Name</th>
        <th>L-T-P</th>
        <th>Credits</th>
        <th>Faculty Name</th>
        <th>Assisting Faculty</th>
    </tr>
    <tr>
        <td>23MAT116</td>
        <td>Discrete Mathematics</td>
        <td>3-0-2</td>
        <td>4</td>
        <td>Dr.Srinivasa Rao Thota</td>
        <td></td>
    </tr>
    <tr>
        <td>23MAT117</td>
        <td>Linear Algebra</td>
        <td>3-0-2</td>
        <td>4</td>
        <td>Mr.Sapavat Bixapathi</td>
        <td></td>
    </tr>
    <tr>
        <td>23CSE111</td>
        <td>Object Oriented Programming</td>
        <td>3-0-2</td>
        <td>4</td>
        <td>Dr.Satya Rajendra Singh R</td>
        <td>Dr.Balaji TK</td>
    </tr>
    <tr>
        <td>23PHY115</td>
        <td>Modern Physics</td>
        <td>2-1-0</td>
        <td>3</td>
        <td>Dr.Soumya</td>
        <td></td>
    </tr>
    <tr>
        <td>23CSE113</td>

```

```

        <td>User Interface Design</td>
        <td>2-0-2</td>
        <td>3</td>
        <td>Dr.Rajkumar Batchu</td>
        <td>Mr.Barge Bharadwaj</td>
    </tr>
    <tr>
        <td>23MEE115</td>
        <td>Manufacturing Practice</td>
        <td>0-0-3</td>
        <td>1</td>
        <td>Dr.Sathies S.</td>
        <td></td>
    </tr>
    <tr>
        <td>22ADM111</td>
        <td>Glimpses of Glorious India</td>
        <td>2-0-1</td>
        <td>2</td>
        <td>Dr.Siva Panuganti</td>
        <td></td>
    </tr>
    <tr>
        <td></td>
        <td>Total(15L+1Tut+12P=28 hrs)</td>
        <td>15-1-12</td>
        <td>21</td>
        <td></td>
        <td></td>
    </tr>
</table>
</body>
</html>

```

Output:

UID-CSE25223 Master Time Table											
Time/ Day	Slot 1 08:55-09:45	Slot 2 09:45-10:35	10:35-10:45	Slot 3 10:45-11:35	Slot 4 11:35-12:25	Slot 5 12:25-01:15	1:15-2:10	Slot 6 02:10-09:45	3:00-3:10	Slot 7 03:10-04:00	Slot 8 04:00-04:50
Monday	23MAT116 Discrete Mathematics_Lab2		Sheet Break	23PHY115 Modern Physics	23CSE111 Object Oriented Programming	22ADM111 Glimpses of Glorious India	Lunch Break	22ADM111 Glimpses of Glorious India(F)	Sheet Break	Library/Sports	
Tuesday	23CSE113 User Interface Design LAB(BYOD)			23MAT117 Linear Algebra	23MAT116 Discrete Mathematics	COUNSELLING		23MEE115 Manufacturing Practice LAB		23MEE115 Manufacturing Practice LAB	
Wednesday	23PHY115 Modern Physics	23CSE113 User Interface Design		23MAT117 Linear Algebra_LAB(BYOD)		COUNSELLING		Library		22ADM111vGlimpses of Glorious India	Library/Sports
Thursday	23CSE111 Object Oriented Programming	23MAT116 Discrete Mathematics		23CSE111Object Oriented Programming_Lab2		23MAT117 Linear Algebra		Library		Library/Sports	
Friday	23CSE111 Object Oriented Programming	23MAT116 Discrete Mathematics		23PHY115 Modern Physics	23PHY115 Modern Physics	23CSE113 User Interface Design		Library		Library/Sports	

Course Code	Course Name	L-T-P	Credits	Faculty Name	Assisting Faculty
23MAT116	Discrete Mathematics	3-0-2	4	Dr.Srinivasa Rao Thota	
23MAT117	Linear Algebra	3-0-2	4	Mr.Sapavat Bikapathi	
23CSE111	Object Oriented Programming	3-0-2	4	Dr.Satya Rajendra Singh R	Dr.Balaji TK
23PHY115	Modern Physics	2-1-0	3	Dr.Soumya	
23CSE113	User Interface Design	2-0-2	3	Dr.Rajkumar Satchu	Mr.Barge Bharadwaj
23MEE115	Manufacturing Practice	0-0-3	1	Dr.Sathies S.	
22ADM111	Glimpses of Glorious India	2-0-1	2	Dr.Siva Panuganti	
Total(15L+1Tut+12P=28 hrs)		15-1-12	21		

B. Aim: To Create an HTML Webpage displaying Syllabus of all the subjects with the timetable.

Code:

```
<!DOCTYPE html>
<html>
  UID-CSE25223
  <body>
    <h1>Syllabus</h1>
    <h2>Discrete Mathematics</h2>
    
    <h2>Linear Algebra</h2>
    
    <h2>Object Oriented Programming</h2>
    
    <h2>Modern Physics</h2>
    
    <h2>Manufacturing Practice Lab</h2>
    
    <h2>User Interface Design</h2>
    
    <h2>Glimpses of Glorius India</h2>
    
    <table border="3" style="font-family: 'Courier New', Courier, monospace;text-align:center;" cellspacing="0" cellpadding="10">
      <tr style="background-color: yellow;">
        <th colspan="12">UID-CSE25223 Master Time Table</th>
      </tr>
```



```

<tr>
  <th rowspan="2">Time/ Day</th>
  <th>Slot 1</th>
  <th>Slot 2</th>
  <th style="background-color: yellow;" rowspan="2">10:35-10:45</th>
  <th>Slot 3</th>
  <th>Slot 4</th>
  <th>Slot 5</th>
  <th style="background-color: yellow;" rowspan="2">1:15-2:10</th>
  <th>Slot 6</th>
  <th style="background-color: yellow;" rowspan="2">3:00-3:10</th>
  <th>Slot 7</th>
  <th>Slot 8</th>
</tr>
<tr>
  <th>08:55-09:45</th>
  <th>09:45-10:35</th>
  <th>10:45-11:35</th>
  <th>11:35-12:25</th>
  <th>12:25-01:15</th>
  <th>02:10-09:45</th>
  <th>03:10-04:00</th>
  <th>04:00-04:50</th>
</tr>
<tr>
  <th>Monday</th>
  <td colspan="2" style="background-color: aqua;">23MAT116 Discrete
Mathematics_lab2</td>
  <th rowspan="5" style="background-color: yellow;vertical-align:bottom;writing-
mode:sideways-lr;">Short Break</th>
  <td>23PHY115 Modern Physics</td>
  <td>23CSE111 Object Oriented Programming</td>
  <td>22ADM111 Glimpses of Glorious India</td>
  <th rowspan="5" style="background-color: yellow;vertical-align:bottom;writing-
mode:sideways-lr;">Lunch Break</th>
  <td>22ADM111 Glimpses of Glorious India(P)</td>
  <th rowspan="5" style="background-color: yellow;vertical-align:bottom;writing-
mode:sideways-lr;">Short Break</th>
  <td colspan="2">Library/Sports</td>
</tr>
<tr>
  <th>Tuesday</th>
  <td colspan="2" style="background-color: rgb(227, 84, 235);">23CSE113 User
Interface Design LAB(BYOD)</td>
  <td>23MAT117 Linear Algebra</td>
  <td>23MAT116 Discrete Mathematics</td>
  <td style="color: red;">COUNSELLING</td>
  <td style="background-color: bisque;">23MEE115 Manufacturing Practice LAB</td>

```

```

        <td colspan="2" style="background-color: bisque;">23MEE115 Manufacturing
Practice LAB</td>
    </tr>
    <tr>
        <th>Wednesday</th>
        <td >23PHY115 Modern Physics</td>
        <td>23CSE113 User Interface Design</td>
        <td colspan="2" style="background-color: aquamarine;">23MAT117 Linear
Algebra_LAB(BYOD)</td>
        <td style="color: red;">COUNSELLING</td>
        <td >Library</td>
        <td>22ADM111vGlimpses of Glorious India</td>
        <td>Library/Sports</td>
    </tr>
    <tr>
        <th>Thursday</th>
        <td >23CSE111 Object Oriented Programming</td>
        <td>23MAT116 Discrete Mathematics</td>
        <td colspan="2" style="background-color: blueviolet;">23CSE111Object Oriented
Programming _Lab2</td>
        <td >23MAT117 Linear Algebra</td>
        <td >Library</td>
        <td colspan="2">Library/Sports</td>
    </tr>
    <tr>
        <th>Friday</th>
        <td >23CSE111 Object Oriented Programming</td>
        <td>23MAT116 Discrete Mathematics</td>
        <td >23PHY115 Modern Physics</td>
        <td >23PHY115 Modern Physics</td>
        <td >23CSE113 User Interface Design</td>
        <td >Library</td>
        <td colspan="2">Library/Sports</td>
    </tr>
</table>
<p><br><br><br><br></p>
<table border="3" style="font-family: 'Courier New', Courier, monospace;"
cellspacing="0" cellpadding="10">
    <tr>
        <th>Course Code</th>
        <th>Course Name</th>
        <th>L-T-P</th>
        <th>Credits</th>
        <th>Faculty Name</th>
        <th>Assisting Faculty</th>
    </tr>
    <tr>
        <td>23MAT116</td>

```

<td><td>Discrete Mathematics</td></td>	<td>Discrete Mathematics</td>
<td><td>3-0-2</td></td>	<td>3-0-2</td>
<td><td>4</td></td>	<td>4</td>
<td><td>Dr.Srinivasa Rao Thota</td></td>	<td>Dr.Srinivasa Rao Thota</td>
<td><td></td></td>	<td></td>

<td><td>23MAT117</td></td>	<td>23MAT117</td>
<td><td>Linear Algebra</td></td>	<td>Linear Algebra</td>
<td><td>3-0-2</td></td>	<td>3-0-2</td>
<td><td>4</td></td>	<td>4</td>
<td><td>Mr.Sapavat Bixapathi</td></td>	<td>Mr.Sapavat Bixapathi</td>
<td><td></td></td>	<td></td>

<td><td>23CSE111</td></td>	<td>23CSE111</td>
<td><td>Object Oriented Programming</td></td>	<td>Object Oriented Programming</td>
<td><td>3-0-2</td></td>	<td>3-0-2</td>
<td><td>4</td></td>	<td>4</td>
<td><td>Dr.Satya Rajendra Singh R</td></td>	<td>Dr.Satya Rajendra Singh R</td>
<td><td>Dr.Balaji TK</td></td>	<td>Dr.Balaji TK</td>

<td><td>23PHY115</td></td>	<td>23PHY115</td>
<td><td>Modern Physics</td></td>	<td>Modern Physics</td>
<td><td>2-1-0</td></td>	<td>2-1-0</td>
<td><td>3</td></td>	<td>3</td>
<td><td>Dr.Soumya</td></td>	<td>Dr.Soumya</td>
<td><td></td></td>	<td></td>

<td><td>23CSE113</td></td>	<td>23CSE113</td>
<td><td>User Interface Design</td></td>	<td>User Interface Design</td>
<td><td>2-0-2</td></td>	<td>2-0-2</td>
<td><td>3</td></td>	<td>3</td>
<td><td>Dr.Rajkumar Batchu</td></td>	<td>Dr.Rajkumar Batchu</td>
<td><td>Mr.Barge Bharadwaj</td></td>	<td>Mr.Barge Bharadwaj</td>

<td><td>23MEE115</td></td>	<td>23MEE115</td>
<td><td>Manufacturing Practice</td></td>	<td>Manufacturing Practice</td>
<td><td>0-0-3</td></td>	<td>0-0-3</td>
<td><td>1</td></td>	<td>1</td>
<td><td>Dr.Sathies S.</td></td>	<td>Dr.Sathies S.</td>
<td><td></td></td>	<td></td>

<td><td>22ADM111</td></td>	<td>22ADM111</td>
--	-------------------

```
<td>Glimpses of Glorius India</td>
<td>2-0-1</td>
<td>2</td>
<td>Dr. Siva Panuganti</td>
<td></td>
</tr>
<tr>
<td></td>
<td>Total(15L+1Tut+12P=28 hrs)</td>
<td>15-1-12</td>
<td>21</td>
<td></td>
<td></td>
</tr>
</table>
</body>
</html>
```

Output:

UID-CSE25223

Syllabus

Discrete Mathematics

Unit 1

Logic, Mathematical Reasoning and Counting: Logic, Propositional Equivalence, Predicate and Quantifiers, Theorem Proving, Recursive Definitions, Recursive Algorithms, Basics of Counting, Pigeonhole Principle, Permutation and Combinations.

Unit 2

Relations and Their Properties: Representing Relations, Closure of Relations, Partial Ordering, Equivalence Relations and partitions.

Advanced Counting Techniques and Relations: Recurrence Relations, Solving Recurrence Relations, Generating Functions, Solutions of Homogeneous Recurrence Relations, Divide and Conquer Relations, Inclusion-Exclusion.

Unit 3

Graph Theory: Graphs and Sub graphs, isomorphism, matrices associated with graphs, degrees, walks, connected graphs, shortest path algorithm. Euler and Hamilton Graphs: Euler graphs, Euler's theorem. Fleury's algorithm for Eulerian trails. Hamilton cycles, Chinese-postman problem, approximate solutions of traveling salesman problem. Closest neighbour algorithm.

Linear Algebra

Unit 1

Vector Spaces: Vector spaces - Sub spaces - Linear independence - Basis - Dimension; Inner Product Spaces: Inner products - Orthogonality - Orthogonal basis - Gram Schmidt Process - Change of basis - Orthogonal complements - Projection on subspace - Least Square Principle. QR- Decomposition.

Unit 2

Linear Transformations: Linear transformation - Relation between matrices and linear transformations - Kernel and range of a linear transformation - Change of basis - Nilpotent transformations. Symmetric and Skew Symmetric Matrices, Adjoint and Hermitian Adjoint of a Matrix, Hermitian, Unitary and Normal Transformations, Self-Adjoint and Normal Transformations.

Unit 3

Eigen values and Eigen vectors: Eigen Values and Eigen Vectors, Diagonalization, Orthogonal Diagonalization, Quadratic Forms, Diagonalizing Quadratic Forms, Conic Sections. Similarity of linear transformations - Diagonalization and its applications - Jordan form and rational canonical form. Case Studies: Applications on least square and image transformations.

Object Oriented Programming

Unit1

Structured to Object Oriented Approach by Examples - Object Oriented languages - Properties of Object Oriented system - UML and Object Oriented Software Development - Use case diagrams and documents as a functional model - Identifying Objects and classes - Representation of Objects and its state by Object Diagram - Simple Class using class diagram - Encapsulation - Data Hiding - Reading and Writing Objects - Class Level and Instance Level Attributes and Methods- JIVE environment for debugging.

Unit2

Aggregation and Composition using Class Diagram - Generalization using Class Diagram - Inheritance - Constructor and Over Riding - Visibility - Attribute - Parameter - Package - Local and Global - Polymorphism - Overloading - Abstract Classes and Interfaces.

Unit3

Exception Handling - Inner Classes - Wrapper classes - String - and String Builder classes - Number - Math - Random - Array methods - File Streams - Serialization - Generics - Collection framework - Comparator and Comparable - Vector and ArrayList - Iterator and Iterable.

Modern Physics

Unit 1

Origin of quantum theory of radiation: Black body radiation, photo-electric effect, Compton Effect – pair production and annihilation, De-Broglie hypothesis, description of waves and wave packets, group velocities. Evidence for wave nature of particles: Davisson-Germer experiment, Heisenberg uncertainty principle.

Unit 2

Atomic structure: Historical Development of atomic structures: Thomson's Model, Rutherford's Model: Scattering formula and its predictions, Atomic spectra - Bohr's Model, Sommerfield's Model, The correspondence principle, nuclear motion, and atomic excitation, Application: Lasers.

Unit 3

Quantum mechanics: Wave function, Probability density, expectation values - Schrodinger equation – time dependent and independent, Linearity and superposition, expectation values, operators, Eigen functions and Eigen values

Unit 4

Application of 1D Schrodinger Wave equation: Free particle, Particle in a box, Finite potential well, Tunnel effect, Harmonic oscillator.

Unit 5

Intro to Quantum computing- Q bits- II Quantum correlations: Bell inequalities and entanglement, Schmidt decomposition, super dense coding, teleportation. Module

Manufacturing Practice Lab

1. Additive Manufacturing Laboratory –12 hours

Introduction to digital manufacturing. Introduction to Additive Manufacturing - types – additive manufacturing applications - Materials for 3D printing, CAD Modelling for Additive manufacturing, Slicing and STL file generation - G code generation - 3D printing of simple geometries.

2. Mechanical Engineering Laboratory –12 hours

Study of tools and equipment used for basic manufacturing processes.
Manual arc welding practice for making Butt and Lap joints - Soldering Practice
Introduction to Machine Tools and Machining Processes

3. Automation lab –12 hours

Design, simulate and test pneumatic and electro-pneumatic circuits. Introduction to PLC –PLC programming for automation applications.

4. Automobile Engineering lab –9 hours

Overview of automobiles – components –functioning of various sub-systems; Power train, steering system, suspension system and braking system. Introduction to electric vehicles, hybrid vehicles, alternate fuels. Introduction to E Mobility.

User Interface Design

Unit 1

Introduction to Internet and Web design – Working of Web – Roles of Front-end, Back-end and Full stack development – Web Server – Internet protocols – Web Hosting – HTML – HTML Tags – Create a simple Web Page – Marking up Text – Adding Links – Adding Images – SVG – Table Markup – Embedded Media - Web Based Forms – HTML Forms - CSS for Presentation - Basic Style Sheet - A CSS Style Primer - Using Style Classes - Using Style IDs - Formatting Text - Advanced Typography with CSS3 - Working with Margins, Padding, Alignment, and Floating. Responsive web design, Introduction to version control systems.

Unit 2

CSS Box Model and Positioning - Creating Layouts Using Modern CSS - Backgrounds and Borders - CSS Transformations and Transitions - Animating with CSS and the Canvas - Dynamic Web Pages – Web Scripting - JavaScript programming for Web Applications – Including JavaScript in HTML – HTML5 Form Controls and Validation - Document Object Model - DOM manipulation and events - Event Handlers

Unit 3

Overview of UI Frameworks – User Interface Design Concepts in Web Development – Accessibility - Introduction to Electron - Overview of Electron - Setting up an Electron project - Creating a basic desktop application with Electron - Object communication - Introduction to other popular frameworks and libraries (e.g., JavaFX, WPF, GTK).

Glimpses of Glorius India

Face the Brutes, Role of Women in India, Acharya Chanakya, God and Isvara, Bhagavad Gita: From Soldier to Samsarin to Sathika, Lessons of Yoga from Bhagavad Gita, Indian Soft powers, Preserving Nature through Faith, Ancient Indian Cultures (Class Activity), Practical Vedanta, To the World from India, Indian Approach to Science.

UD-CSE2523 Master Time Table									
Time/ Day	Slot 1	Slot 2	Slot 3	Slot 4	Slot 5	Slot 6	Slot 7	Slot 8	
	08:55-09:45	09:45-10:35	10:45-11:35	11:35-12:25	12:25-01:15	02:10-09:45	03:10-04:00	04:00-04:50	
Monday	23CSE116 Discrete Mathematics_Lab2	Lunch Break	23PHY115 Modern Physics	23CSE111 Object Oriented Programming	22ADM11 Glimpses of Glorious India	Lunch Break	Break		
Tuesday	23CSE113 User Interface Design_LAB (BYOD)		23CSE117 Linear Algebra	23CSE116 Discrete Mathematics	COUNSELLING		23CSE115 Manufacturing Practice LAB	Library/Sports	
Wednesday	23PHY115 Modern Physics		23CSE117 Linear Algebra_LAB (BYOD)		COUNSELLING		23CSE115 Manufacturing Practice LAB	Library/Sports	
Thursday	23CSE111 Object Oriented Programming		23CSE110 Object Oriented Programming_Lab2		23MAT117 Linear Algebra		23CSE115 Manufacturing Practice LAB	Library/Sports	
Friday	23CSE111 Object Oriented Programming		23PHY115 Modern Physics	23PHY115 Modern Physics	23CSE113 User Interface Design		Library/Sports	Library/Sports	

Course Code	Course Name	L-T-P	Credits	Faculty Name	Assisting Faculty
23MAT116	Discrete Mathematics	3-0-2	4	Dr. Srinivas Rao Thota	
23MAT117	Linear Algebra	3-0-2	4	Mr. Agavut Sivagathi	
23CSE111	Object Oriented Programming	3-0-2	4	Dr. Savya Rajendra Singh R	Dr. Balaji TK
23PHY115	Modern Physics	2-1-0	3	Dr. Soumya	
23CSE113	User Interface Design	2-0-2	3	Dr. Rajkumar Raschu	Mr. Barge Bharadwaj
23CSE115	Manufacturing Practice	0-0-3	1	Dr. Sathies S.	
22ADM11	Glimpses of Glorius India	2-0-1	2	Dr. Siva Panuganti	
	Total (15S+1Tus+12P=28 hrs)	15-1-12	21		

Explanation of the Tags:

- **rowspan** - rowspan="n" – Merges n cells vertically
- **colspan** - colspan="n" – Merges n cells horizontally
- **cellspacing** - cellspacing="m" - Space between table cells
- **cellpadding** - cellpadding="m" - Padding inside table cells
- **background-color** - background-color: aqua; - Cell background color to aqua
- **color** - color: red; - Text color for counselling
- **vertical-align** - vertical-align: bottom; - Bottom cell alignment
- **writing-mode** - writing-mode: sideways-lr; - Rotates break labels
- **font-family** - font-family: 'Courier New', monospace; - Monospace font
- **text-align** - text-align: center; - Center the table content

WEEK-4

4.A. **Aim:** To collect student details through a structured HTML form with proper labels and table layout.

Code:

```
<!DOCTYPE html>
<html>
  UID-CSE25223
  <body>
    <p>P.Akhil</p>
    <h1 align="center">HTML Forms</h1>
    <table border="1" cellpadding="10" cellspacing="0" align="center">
      <tr>
        <th>Full Name</th>
        <td><input type="text" name="fname"></td>
      </tr>
      <tr>
        <th>Email</th>
        <td><input type="email" name="email"></td>
      </tr>
      <tr>
        <th>Phone Number</th>
        <td><input type="tel" name="phone"></td>
      </tr>
      <tr>
        <th>Gender</th>
        <td>
          <input type="radio" name="gender" value="male"> Male
          <input type="radio" name="gender" value="female"> Female
          <input type="radio" name="gender" value="other"> Other
        </td>
      </tr>
      <tr>
        <th>Course Applying for?</th>
        <td>
          <select name="course">
            <option value="bt">BTech</option>
            <option value="bba">BBA</option>
            <option value="mbbs">MBBS</option>
          </select>
        </td>
      </tr>
      <tr>
        <th>Hobbies</th>
        <td>
          <input type="checkbox" name="hobby" value="sports"> Sports
        </td>
      </tr>
    </table>
  </body>
</html>
```

```

        <input type="checkbox" name="hobby" value="music"> Music
        <input type="checkbox" name="hobby" value="reading"> Reading
    </td>
</tr>
<tr>
    <th>Date of Birth</th>
    <td><input type="date" name="dob"></td>
</tr>
<tr>
    <th>Upload Certificate</th>
    <td><input type="file" name="certi"></td>
</tr>
<tr>
    <th>Address</th>
    <td><textarea name="address" rows="4" cols="50"></textarea></td>
</tr>
<tr>
    <td colspan="2" align="center"><input type="submit" value="Submit"><input
type="reset" value="Reset"></td>
</tr>
</table>
</body>

```

Output:

UID-CSE25223

P.Akhil

HTML Forms

Full Name	
Email	
Phone Number	
Gender	☐ Male ☐ Female ☐ Other
Course Applying for?	BTech v
Hobbies	☐ Sports ☐ Music ☐ Reading
Date of Birth	mm / dd / yyyy v
Upload Certificate	Choose File No file chosen
Address	
Submit Reset	

Explanation of the tags:

- `<input type="text">`: Creates a single-line text box for basic input like names.
- `<input type="email">`: Makes a text field that expects and validates email format.
- `<input type="tel">`: Input field designed for telephone numbers.
- `<input type="radio">`: Lets users pick **one** option from a group (like gender).
- `<select>`: Creates a **dropdown** menu with predefined choices.
- `<option>`: Individual items inside a dropdown/select menu.
- `<input type="checkbox">`: Lets users select **multiple** options (like hobbies).
- `<input type="date">`: Opens a calendar picker for selecting dates.
- `<input type="file">`: Button to upload and select files from computer.
- `<textarea>`: Multi-line text input box for longer content like addresses.
- `<input type="submit">`: Button that sends form data when clicked.
- `<input type="reset">`: Button that clears all form fields.
- `colspan="2"`: Makes one cell stretch across **two columns** in the table.

4.B. **Aim:** To create a webpage showing Login and Registration forms side-by-side using iframes for easy embedding.

Code:

Login.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Login Form</title>
</head>
<body style="background-color: antiquewhite; font-family: Arial; text-align: center;">

  <div class="box">
    <h3>User Login</h3>

    <form >
      <table style="text-align: center;" align="center" cellpadding="10"
cellspacing="0" border="1">
        <tr>
          <td>Username:</td>
          <td><input type="text" name="username"></td>
        </tr>
        <tr></tr>
          <td>Password:</td>
          <td><input type="password" name="password"></td>
        </tr>
        <tr>
          <td colspan="2" align="center"><button type="submit">Login</button></td>
        </tr>
      </table>
    </form>
  </div>

</body>
</html>
```

Register.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Registration Form</title>
</head>
<body style="background-color: antiquewhite; font-family: Arial; text-align: center;">

  <div class="box">
    <h3>User Registration</h3>
```



```

    }

    h2 {
        margin-top: 20px;
    }

    .container {
        display: flex;
        justify-content: center;
        gap: 40px;
        margin-top: 30px;
    }

    iframe {
        width: 350px;
        height: 400px;
        border: 2px solid #555;
        background-color: #fff;
    }
</style>
</head>
<body>

    <h2>Two Forms Displayed Using Iframes</h2>

    <div class="container">
        <iframe src="login.html" name="loginFrame"></iframe>
        <iframe src="register.html" name="registerFrame"></iframe>
    </div>

</body>
</html>

```

Output:

UID-CSE25223

Two Forms Displayed Using Iframes

User Login

Username:	
Password:	
Login	

User Registration

Field	Input
Full Name:	
Email:	
Phone Number:	
Password:	
Register Reset	

Explanation of the tags:

- `<iframe>`: Embeds another HTML page (like login.html) inside the current webpage as a frame.
- `<div class="box">`: Groups content together and applies CSS styling through the class name.
- `<button type="submit">`: Styled clickable button that submits form data (looks better than basic input).
- `<th>`: Makes table headers bold and centered (used in registration form headers).
- `style=""`: Inline CSS that styles elements directly (background-color, font-family, text-align).
- `.container`: CSS class using flex layout to position two iframes side by side.
- `display: flex`: CSS property that arranges items horizontally in a row.
- `justify-content: center`: CSS centers the flex items (iframes) horizontally.
- `gap: 40px`: CSS adds space between the two side-by-side iframes.

WEEK-5

5.A. **Aim:** To create a card layout using <div> tag consider three cards and a button using HTML and CSS

Code:

```
UID-AV.SC.U4CSE25223
<!DOCTYPE html>
<html>
<head>
  <title>Indian Freedom Fighters</title>
  <style>
    body {
      font-family: Arial;
      background: #0f172a;
      text-align: center;
    }
    h1 {
      background: #3fd9a5;
      padding: 20px;
      color: white;
    }
    .container {
      display: flex;
      flex-wrap: wrap;
      justify-content: center;
      gap: 20px;
      padding: 30px;
    }
    .card {
      background: white;
      color: black;
      width: 260px;
      padding: 20px;
      border-radius: 12px;
      box-shadow: 0 5px 15px gray;
    }
    .hero-img {
      width: 150px;
      height: 200px;
      object-fit: cover;
      border-radius: 8px;
    }
    button {
      padding: 8px 16px;
      border: none;
      background: #3fd9a5;
```



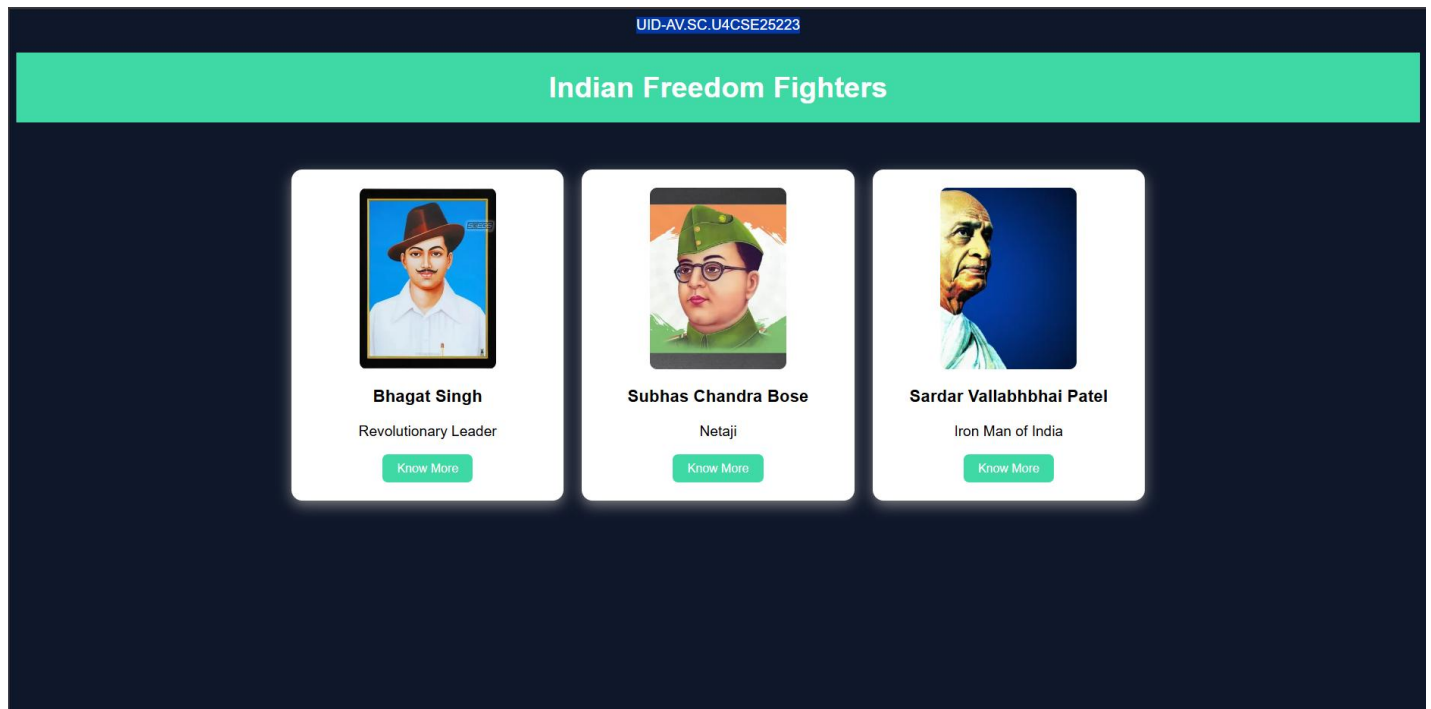
```
    color: white;
    border-radius: 6px;
    cursor: pointer;
  }
  button:hover {
    background: black;
  }
</style>
</head>
<body>
  <h1>Indian Freedom Fighters</h1>

  <div class="container">
    <div class="card">
      
      <h3>Bhagat Singh</h3>
      <p>Revolutionary Leader</p>
      <button>Know More</button>
    </div>

    <div class="card">
      
      <h3>Subhas Chandra Bose</h3>
      <p>Netaji</p>
      <button>Know More</button>
    </div>

    <div class="card">
      
      <h3>Sardar Vallabhbhai Patel</h3>
      <p>Iron Man of India</p>
      <button>Know More</button>
    </div>
  </div>
</body>
</html>
```

Output:



Explanation of the tags:

- `.container { ... }`: CSS class that uses flexbox to position cards side by side with spacing.
- `display: flex`: Lays out card items in a flexible row/line.
- `flex-wrap: wrap`: Allows cards to move to next line on smaller screens.
- `justify-content: center`: Centers all cards horizontally in the container.
- `gap: 20px`: Adds consistent space between each card.
- `.card { ... }`: Styles individual card boxes with size, shadow, rounded corners.
- `border-radius: 12px`: Rounds card corners for modern look.
- `box-shadow: 0 5px 15px gray`: Creates drop shadow to make cards appear lifted.
- `.hero-img { ... }`: Styles images inside cards with fixed dimensions.
- `object-fit: cover`: Crops images to fill container without distortion.
- `button:hover { ... }`: Changes button appearance when mouse hovers over it.

5.B. **Aim:** Navigation using iframe (mini website) upon clicking each buttons the details about that button should be display on the same page

Code:

Main File:

```
UID-AV.SC.U4CSE25223
<!DOCTYPE html>
<html>
<head>
<title>Bhagat Singh</title>
<style>
body{margin:0;font-family:Arial;background-color: gray;}
.menu{
  background:#b91c1c;
  padding:15px;
  text-align:center;
}
.menu a{
  color:white;
  margin:0 15px;
  text-decoration:none;
  background:black;
  padding:8px 12px;
}
iframe{
  width:100%;
  height:400px;
  border:none;
}
</style>
</head>
<body>
<h1 style="text-align:center;background:#b91c1c;color:white;padding:20px;">
Bhagat Singh
</h1>

<div class="menu">
  <a href="strengths.html" target="frame">Ideals & Weaknesses</a>
  <a href="personal.html" target="frame">Personal Info</a>
  <a href="contribution.html" target="frame">Contributions</a>
</div>

<iframe name="frame"></iframe>
</body>
</html>
```

strengths.html:

```
<!DOCTYPE html>
<html>
<head>
<style>
body{font-family:Arial;text-align:center;}
.box{
  border:2px solid black;
  padding:20px;
  display:inline-block;
}
h2{color:#b91c1c;}
</style>
</head>
<body>
<div class="box">
  <h2>Personal Information</h2>
  <p>Name: Bhagat Singh</p>
  <p>Born: 28 September 1907, Punjab</p>
  <p>Movement: Indian Independence Movement</p>
  
</div>
</body>
</html>
```

personal.html:

```
<!DOCTYPE html>
<html>
<head>
<style>
body{font-family:Arial;text-align:center;}
.box{
  border:2px solid black;
  padding:20px;
  display:inline-block;
}
h2{color:#b91c1c;}
</style>
</head>
<body>
<div class="box">
  <h2>Personal Information</h2>
  <p>Name: Bhagat Singh</p>
  <p>Born: 28 September 1907, Punjab</p>
```

```
<p>Movement: Indian Independence Movement</p>

</div>
</body>
</html>
```

contribution.html:

```
<!DOCTYPE html>
<html>
<head>
<style>
body{font-family:Arial;text-align:center;}
.box{
  border:2px solid black;
  padding:20px;
  display:inline-block;
}
h2{color:#8edc7b;}
</style>
</head>
<body>
<div class="box">
  <h2>Contributions</h2>
  <p>Inspired youth to join the freedom struggle.</p>
  <p>Protested against British rule through revolutionary activities.</p>
  <p>Became a symbol of sacrifice after his martyrdom.</p>
  
</div>
</body>
</html>
```

Output:

UID-AV.SC.U4CSE25223

Bhagat Singh

Ideals & Weaknesses

Personal Info

Contributions

Ideals and Weaknesses

Ideals: Courage, patriotism, sacrifice for freedom.

Weaknesses: Impulsive decisions, readiness for extreme risk.



file:///C:/Users/asus/Documents/UID_CSE25223/strengths.html

UID-AV.SC.U4CSE25223

Bhagat Singh

Ideals & Weaknesses

Personal Info

Contributions

Personal Information

Name: Bhagat Singh

Born: 28 September 1907, Punjab

Movement: Indian Independence Movement



file:///C:/Users/asus/Documents/UID_CSE25223/personal.html

UID-AV.SC.U4CSE25223

Bhagat Singh

Ideals & Weaknesses

Personal Info

Contributions

Contributions

Inspired youth to join the freedom struggle.

Protested against British rule through revolutionary activities.

Became a symbol of sacrifice after his martyrdom.



Explanation of the tags:

- `iframe { ... }`: Styles embedded frame (full width, 400px height, no border).
- `target="frame"`: Loads linked pages inside specific iframe by name.
- `name="frame"`: Names iframe so menu links can target it.
- `.menu a { ... }`: Styles navigation links (spacing, no underline, black background).
- `margin:0 15px`: Horizontal spacing between menu links.
- `text-decoration:none`: Removes underline from clickable links.
- `.box { ... }`: Styles sub-page content box (border, padding, inline-block).
- `display:inline-block`: Box fits content width, sits in line.

Main File:

```
UID-AV.SC.U4CSE25223
<!DOCTYPE html>
<html>
<head>
  <title>Amrita Vishwa Vidyapeetham</title>
  <style>
    body {
      margin: 0;
      font-family: Arial;
      background: rgb(50, 50, 100);
    }
    header {
      background: #ff4444;
      color: #fff;
      text-align: center;
      padding: 20px;
      font-size: 36px;
    }
    nav {
      background: #0066cc;
      text-align: center;
      padding: 10px;
    }
    nav a {
      color: #ffff00;
      margin: 12px;
      text-decoration: none;
      font-weight: bold;
      padding: 5px;
    }
    nav a:hover {
      background: #fff;
      color: #0066cc;
      border-radius: 5px;
    }
    main {
      display: flex;
      padding: 20px;
    }
    .login {
      width: 350px;
      background: #ffaa00;
      padding: 15px;
      border-radius: 10px;
      color: white;
    }
    .login input[type="email"],
```



```

.login input[type="password"] {
  width: 90%;
  padding: 6px;
  margin: 6px 0;
}
.remember {
  display: flex;
  align-items: center;
  gap: 6px;
  margin: 8px 0;
}
iframe {
  flex: 1;
  margin-left: 20px;
  height: 520px;
  border: none;
  background: #aaffff;
  border-radius: 10px;
}
</style>
</head>
<body>
  <header>AMRITA VISHWA VIDHYAPEETAM</header>

  <nav>
    <a href="amaravati.html" target="campus">AMARAVATI</a>
    <a href="amritapuri.html" target="campus">AMRITAPURI</a>
    <a href="bengaluru.html" target="campus">BENGALURU</a>
    <a href="chennai.html" target="campus">CHENNAI</a>
    <a href="coimbatore.html" target="campus">COIMBATORE</a>
    <a href="faridabad.html" target="campus">FARIDABAD</a>
    <a href="kochi.html" target="campus">KOCHI</a>
    <a href="mysuru.html" target="campus">MYSURU</a>
  </nav>

  <main>
    <div class="login">
      <h2>Login</h2>
      Email: <input type="email">
      Password: <input type="password">
      <div class="remember">
        <input type="checkbox">
        <label>Save me</label>
      </div>
      <input type="button" value="Login" style="width:100%;padding:6px;">
      <br><br>
      <a href="#">Forgot Password?</a><br>
      No account? <a href="#">Register</a>
    </div>
  </main>
</body>
</html>

```

```
</div>

<iframe name="campus" src="amaravati.html"></iframe>
</main>
</body>
</html>
```

amaravati.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Amaravati Campus</title>
</head>
<body style="font-family:Arial;text-align:center;padding:30px;">
  <h1 style="color:#ff4444;">Amaravati Campus</h1>
  <p>
    Amrita Amaravati campus is located near the Krishna River in Andhra Pradesh.
    It provides modern classrooms, research labs, and excellent hostel facilities.
  </p>
  <p>
    Popular Courses: CSE, AI, Data Science<br>
    Facilities: Library, Smart Classrooms, Hostel<br>
    Focus: Innovation and Research
  </p>
  
</body>
</html>
```

amritapuri.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Amritapuri Campus</title>
</head>
<body style="font-family:Arial;text-align:center;padding:30px;">
  <h1 style="color:#ff4444;">Amritapuri Campus</h1>
  <p>
    Amritapuri campus in Kerala is the main campus of Amrita University,
    located near the Arabian Sea with a peaceful academic atmosphere.
  </p>
  <p>
    Popular Courses: Engineering, Arts, Sciences<br>
    Facilities: Research Labs, Beach Campus, Hostel<br>
    Focus: Academic Excellence with Values
  </p>
</body>
</html>
```

```
</p>

</body>
</html>
```

bengaluru.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Bengaluru Campus</title>
</head>
<body style="font-family:Arial;text-align:center;padding:30px;">
  <h1 style="color:#ff4444;">Bengaluru Campus</h1>
  <p>
    Located in India's tech hub, Bengaluru campus offers advanced engineering programs
    with strong industry connections.
  </p>
  <p>
    Popular Courses: Robotics, CSE, ECE<br>
    Facilities: Coding Labs, Innovation Centres<br>
    Focus: Industry Collaboration
  </p>
  
</body>
</html>
```

Chennai.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Chennai Campus</title>
</head>
<body style="font-family:Arial;text-align:center;padding:30px;">
  <h1 style="color:#ff4444;">Chennai Campus</h1>
  <p>
    Chennai campus provides top-quality computing and management education
    with modern infrastructure and placement support.
  </p>
  <p>
    Popular Courses: IT, MBA, Electronics<br>
    Facilities: Digital Library, Placement Cell<br>
    Focus: Career Development
  </p>
  
</body>
```

```
</html>
```

Coimbatore.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Coimbatore Campus</title>
</head>
<body style="font-family:Arial;text-align:center;padding:30px;">
  <h1 style="color:#ff4444;">Coimbatore Campus</h1>
  <p>
    Coimbatore campus is well known for its engineering excellence and
    research-oriented education.
  </p>
  <p>
    Popular Courses: Mechanical, Civil, CSE<br>
    Facilities: Workshops, Labs, Hostel<br>
    Focus: Core Engineering Skills
  </p>
  
</body>
</html>
```

Faridabad.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Faridabad Campus</title>
</head>
<body style="font-family:Arial;text-align:center;padding:30px;">
  <h1 style="color:#ff4444;">Faridabad Campus</h1>
  <p>
    Faridabad campus focuses on medical and healthcare education with
    advanced hospital training facilities.
  </p>
  <p>
    Popular Courses: Medical, Nursing, Allied Health<br>
    Facilities: Teaching Hospital, Labs<br>
    Focus: Healthcare Research
  </p>
  
</body>
</html>
```

Kochi.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Kochi Campus</title>
</head>
<body style="font-family:Arial;text-align:center;padding:30px;">
  <h1 style="color:#ff4444;">Kochi Campus</h1>
  <p>
    Kochi campus offers programs in business, commerce, and applied sciences
    with modern facilities.
  </p>
  <p>
    Popular Courses: MBA, Commerce, Finance<br>
    Facilities: Seminar Halls, Library<br>
    Focus: Industry Training
  </p>
  
</body>
</html>
```

Mysuru.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Mysuru Campus</title>
</head>
<body style="font-family:Arial;text-align:center;padding:30px;">
  <h1 style="color:#ff4444;">Mysuru Campus</h1>
  <p>
    Mysuru campus is known for its peaceful environment, green campus,
    and quality education.
  </p>
  <p>
    Popular Courses: Arts, Sciences, Engineering<br>
    Facilities: Hostel, Library, Sports Complex<br>
    Focus: Calm Learning Atmosphere
  </p>
  
</body>
</html>
```

Output:

UID-AV.SC.U4CSE25223

AMRITA VISHWA VIDHYAPEETAM

AMARAVATIAMRITAPURIBENGALURUCHENNAICOIMBATOREFARIDABADKOCHIMYSURU

Login

Email:

Password:

Save me

Login

Forgot Password?

No account? Register

Amaravati Campus

Amrita Amaravati campus is located near the Krishna River in Andhra Pradesh. It provides modern classrooms, research labs, and excellent hostel facilities.

Popular Courses: CSE, AI, Data Science

Facilities: Library, Smart Classrooms, Hostel

Focus: Innovation and Research



file:///C:/Users/assu/Documents/UID_CSE25223/amaravati.html

UID-AV.SC.U4CSE25223

AMRITA VISHWA VIDHYAPEETAM

AMARAVATIAMRITAPURIBENGALURUCHENNAICOIMBATOREFARIDABADKOCHIMYSURU

Login

Email:

Password:

Save me

Login

Forgot Password?

No account? Register

Amritapuri Campus

Amritapuri campus in Kerala is the main campus of Amrita University, located near the Arabian Sea with a peaceful academic atmosphere.

Popular Courses: Engineering, Arts, Sciences

Facilities: Research Labs, Beach Campus, Hostel

Focus: Academic Excellence with Values



file:///C:/Users/assu/Documents/UID_CSE25223/amritapuri.html

UID-AV.SC.U4CSE25223

AMRITA VISHWA VIDHYAPEETAM

AMARAVATIAMRITAPURIBENGALURUCHENNAICOIMBATOREFARIDABADKOCHIMYSURU

Login

Email:

Password:

Save me

Login

Forgot Password?

No account? Register

Bengaluru Campus

Located in India's tech hub, Bengaluru campus offers advanced engineering programs with strong industry connections.

Popular Courses: Robotics, CSE, ECE

Facilities: Coding Labs, Innovation Centres

Focus: Industry Collaboration



UID-AV/SC.U4CSE25223

AMRITA VISHWA VIDHYAPEETAM

AMARAVATI AMRITAPURI BENGALURU **CHENNAI** COIMBATORE FARIDABAD KOCHI MYSURU

Login

Email:

Password:

☐ Save me

Login

[Forgot Password?](#)

No account? [Register](#)

Chennai Campus

Chennai campus provides top-quality computing and management education with modern infrastructure and placement support.

Popular Courses: IT, MBA, Electronics
Facilities: Digital Library, Placement Cell
Focus: Career Development



file:///C:/Users/ajay/Documents/UID_CSE25223/chennai.html

UID-AV/SC.U4CSE25223

AMRITA VISHWA VIDHYAPEETAM

AMARAVATI AMRITAPURI BENGALURU **CHENNAI** COIMBATORE FARIDABAD KOCHI MYSURU

Login

Email:

Password:

☐ Save me

Login

[Forgot Password?](#)

No account? [Register](#)

Coimbatore Campus

Coimbatore campus is well known for its engineering excellence and research-oriented education.

Popular Courses: Mechanical, Civil, CSE
Facilities: Workshops, Labs, Hostel
Focus: Core Engineering Skills



UID-AV/SC.U4CSE25223

AMRITA VISHWA VIDHYAPEETAM

AMARAVATI AMRITAPURI BENGALURU **CHENNAI** COIMBATORE FARIDABAD KOCHI MYSURU

Login

Email:

Password:

☐ Save me

Login

[Forgot Password?](#)

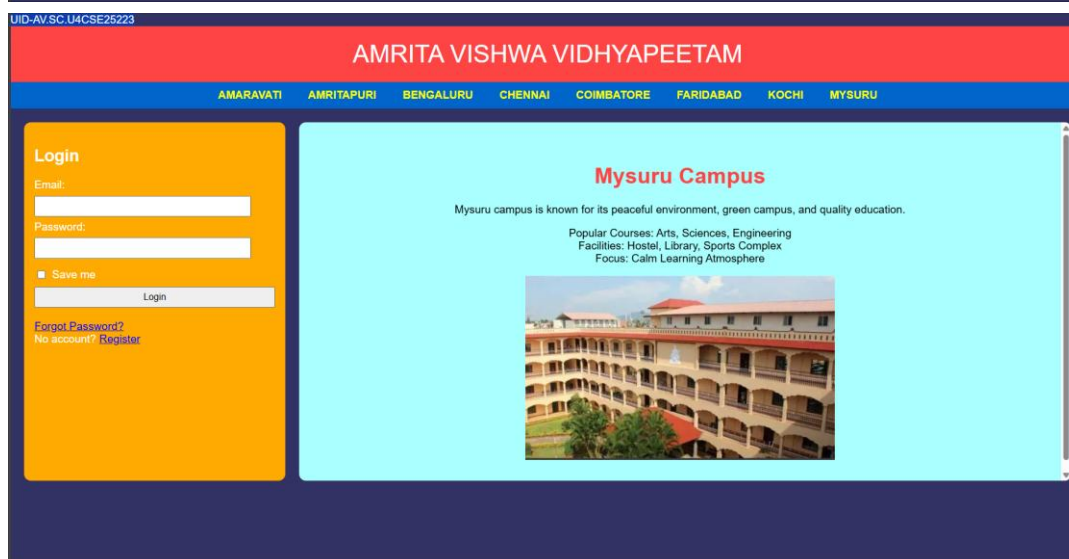
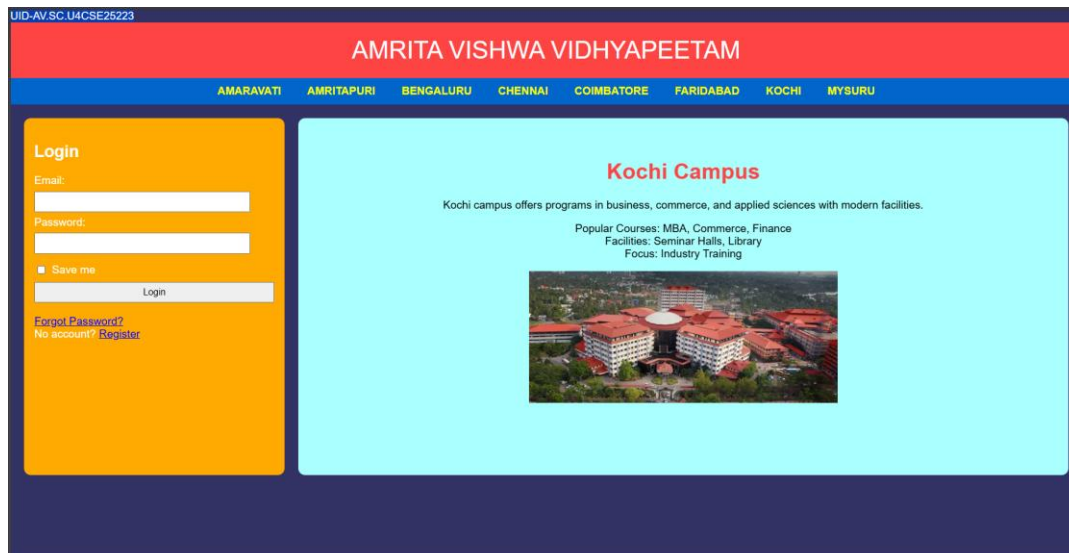
No account? [Register](#)

Faridabad Campus

Faridabad campus focuses on medical and healthcare education with advanced hospital training facilities.

Popular Courses: Medical, Nursing, Allied Health
Facilities: Teaching Hospital, Labs
Focus: Healthcare Research





Explanation of the tags:

- header: Groups main title bar with bold styling (bright red background, large font).
- nav: Creates navigation bar with campus links (brighter blue background, centered).
- nav a: hover: Changes link appearance on mouseover (white background, blue text, rounded).
- main {display: flex}: Places login form and iframe side by side horizontally.
- .login: Styles login box (orange background, rounded corners, white text).
- .remember {display: flex}: Aligns checkbox and "Save me" label in a horizontal row.
- align-items: center: Vertically centers checkbox with its label.

- `gap: 6px`: Adds small space between checkbox and label.
- `iframe {flex: 1}`: Makes iframe take remaining width beside login box.
- `target="campus"`: Loads campus pages inside specific iframe named "campus".
- `name="campus"`: Names iframe so nav links target it specifically.

WEEK-6

6.A. Aim: Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books.

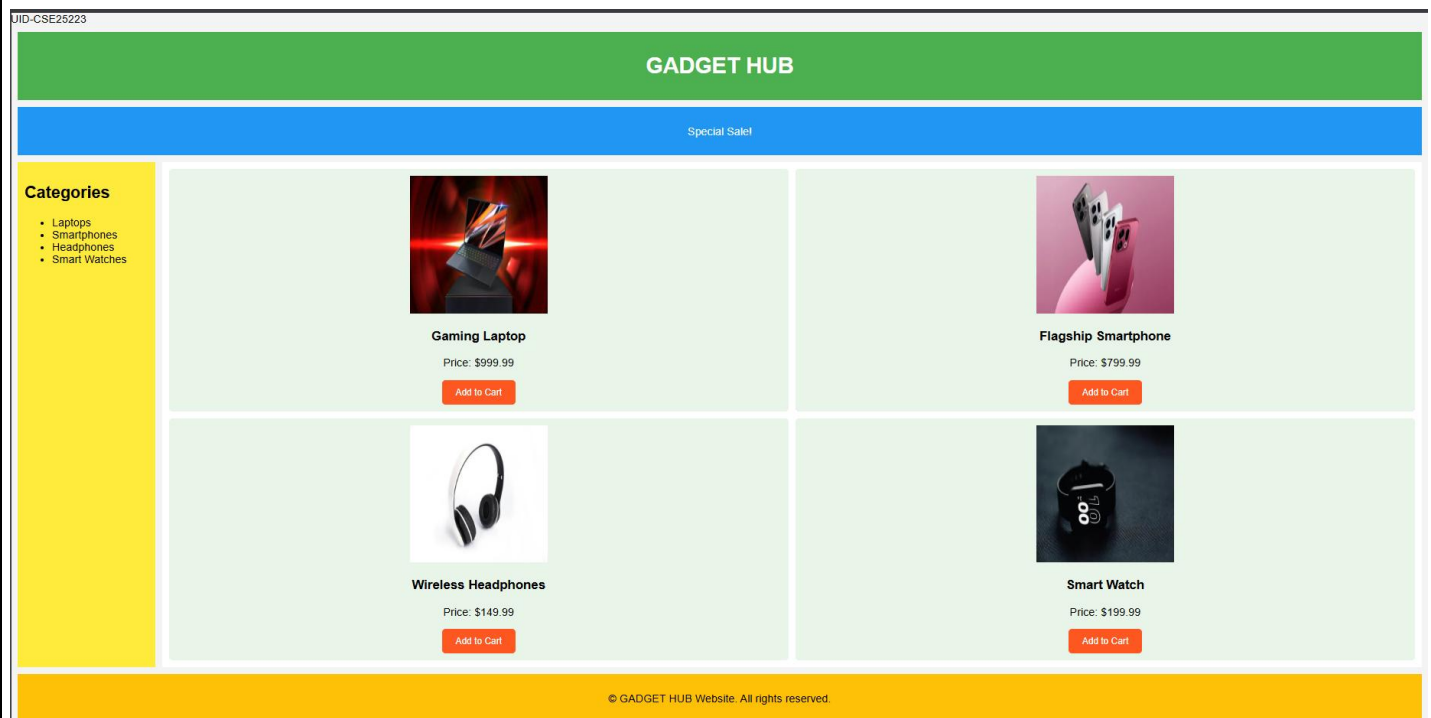
Code:-

```
<!DOCTYPE html>
UID-CSE25223
<html>
<head>
  <title>Gadget Hub</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 0;
      background-color: #f4f4f4;
    }
    .container {
      display: grid;
      grid-template-rows: auto auto 1fr auto;
      grid-template-columns: 200px 1fr;
      grid-template-areas:
        "header header"
        "sale sale"
        "sidebar main"
        "footer footer";
      gap: 10px;
      padding: 10px;
    }
    .header {
      grid-area: header;
      background-color: #4CAF50;
      padding: 10px;
      text-align: center;
      color: white;
    }
    .sale {
      grid-area: sale;
      background-color: #2196F3;
      padding: 10px;
      text-align: center;
      color: white;
    }
    .sidebar {
```

```
        grid-area: sidebar;
        background-color: #FFEB3B;
        padding: 10px;
    }
    .main {
        grid-area: main;
        background-color: white;
        padding: 10px;
        display: grid;
        grid-template-columns: repeat(2, 1fr);
        gap: 10px;
    }
    .footer {
        grid-area: footer;
        background-color: #FFC107;
        padding: 10px;
        text-align: center;
    }
    .product {
        background-color: #e8f5e8;
        padding: 10px;
        border-radius: 5px;
        text-align: center;
    }
    .product-image {
        width: 200px;
        height: 200px;
    }
    .button {
        background-color: #FF5722;
        color: white;
        border: none;
        padding: 10px 20px;
        border-radius: 5px;
    }
    .button:hover {
        background-color: #E64A19;
    }
</style>
</head>
<body>
    <div class="container">
        <div class="header">
            <h1>GADGET HUB</h1>
        </div>
        <div class="sale">
            <p>Special Sale!</p>
        </div>
    </div>
```

```
<div class="sidebar">
  <h2>Categories</h2>
  <ul>
    <li>Laptops</li>
    <li>Smartphones</li>
    <li>Headphones</li>
    <li>Smart Watches</li>
  </ul>
</div>
<div class="main">
  <div class="product">
    
    <h3>Gaming Laptop</h3>
    <p>Price: $999.99</p>
    <button class="button">Add to Cart</button>
  </div>
  <div class="product">
    
    <h3>Flagship Smartphone</h3>
    <p>Price: $799.99</p>
    <button class="button">Add to Cart</button>
  </div>
  <div class="product">
    
    <h3>Wireless Headphones</h3>
    <p>Price: $149.99</p>
    <button class="button">Add to Cart</button>
  </div>
  <div class="product">
    
    <h3>Smart Watch</h3>
    <p>Price: $199.99</p>
    <button class="button">Add to Cart</button>
  </div>
</div>
<div class="footer">
  <p>&copy; GADGET HUB Website. All rights reserved.</p>
</div>
</div>
</body>
</html>
```

Output:-



Explanation of the tags:

- display: grid: Enables CSS Grid layout system
- grid-template-areas: Defines 5 regions (header/sale/sidebar/main/footer)
- grid-area: header: Assigns element to header grid area
- grid-template-columns: 200px 1fr: Fixed 200px sidebar + flexible main
- gap: 10px: Spacing between all grid items
- padding: 10px: Inner spacing for all sections
- background-color: #4CAF50: Green header background
- text-align: center: Centers content in all sections
- repeat(2, 1fr): Creates 2-column product grid in main
- border-radius: 5px: Rounded corners on products/buttons
- .product-image { width: 200px; height: 200px }: Fixed-size product images
- background-color: #FF5722: Orange Add to Cart buttons
- :hover: Button darkens on mouseover (#E64A19)
- border: none: Removes button borders
- 1fr: Flexible units fill available space

6.B. Aim: Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. Based on the category we select corresponding products should be displayed on the sidebar.

Code:-

Main.html

```
<!DOCTYPE html>
<html>
<head>
  <title>GADGET HUB</title>
  <style>
    body {
      margin: 0;
      font-family: Arial;
      color: #333;
      background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
    }
    .container {
      display: grid;
      grid-template:
```

```

        "header header" auto
        "sale sale" auto
        "sidebar main" 1fr
        "footer footer" auto / 220px 1fr;
    gap: 15px;
    padding: 15px;
}
.header, .sale, .sidebar, .main, .footer {
    background: rgba(255,255,255,.95);
    border-radius: 12px;
    padding: 15px;
    box-shadow: 0 4px 20px rgba(0,0,0,0.1);
}
.header {
    grid-area: header;
    text-align: center;
    font-size: 28px;
    letter-spacing: 3px;
    background: linear-gradient(45deg, #4CAF50, #45a049);
    color: white;
}
.sale {
    grid-area: sale;
    overflow: hidden;
    white-space: nowrap;
    height: 40px;
    position: relative;
    background: linear-gradient(45deg, #2196F3, #21CBF3);
    color: white;
    font-weight: bold;
}
.sale p {
    position: absolute;
    font-size: 20px;
    animation: move 10s linear infinite;
}
@keyframes move {
    from { left: -100%; }
    to { left: 100%; }
}
.sidebar {
    grid-area: sidebar;
    background: linear-gradient(135deg, #FFEB3B, #FF9800);
}
.sidebar ul { list-style: none; padding: 0; }
.sidebar li { margin: 15px 0; }
.sidebar a {
    color: #333;

```

```

        text-decoration: none;
        font-size: 18px;
        transition: .3s;
        display: block;
    }
    .sidebar a:hover {
        color: #2196F3;
        padding-left: 8px;
        background: rgba(255,255,255,0.5);
        border-radius: 6px;
    }
    .main {
        grid-area: main;
        background: white;
    }
    iframe {
        width: 100%;
        height: 520px;
        border: 0;
        border-radius: 10px;
        box-shadow: 0 4px 15px rgba(0,0,0,0.1);
    }
    .footer {
        grid-area: footer;
        text-align: center;
        font-size: 14px;
        background: linear-gradient(45deg, #FFC107, #FF9800);
        color: #333;
    }
</style>
</head>
<body>
    <div class="container">
        <div class="header">GADGET HUB</div>
        <div class="sale">
            <p>💧 SPECIAL SALE – UP TO 50% OFF ON GADGETS 💧 </p>
        </div>
        <div class="sidebar">
            <h2>Categories</h2>
            <ul>
                <li><a href="laptops.html" target="contentFrame">LAPTOPS</a></li>
                <li><a href="smartphones.html" target="contentFrame">SMARTPHONES</a></li>
                <li><a href="headphones.html" target="contentFrame">HEADPHONES</a></li>
                <li><a href="smartwatches.html" target="contentFrame">SMART WATCHES</a></li>
            </ul>
        </div>
        <div class="main">
            <iframe name="contentFrame"></iframe>

```



```
    </div>
    <div class="footer">© GADGET HUB Website</div>
  </div>
</body>
</html>
```

laptops.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Laptops</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 20px;
      background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
    }
    .container {
      display: grid;
      grid-template-columns: 1fr 1fr;
      gap: 20px;
    }
    .card {
      background: white;
      padding: 20px;
      border-radius: 15px;
      box-shadow: 0 8px 25px rgba(0,0,0,0.1);
      text-align: center;
    }
    .card-img {
      width: 200px;
      height: 200px;
      border-radius: 12px;
      margin-bottom: 15px;
    }
    .price {
      font-size: 20px;
      color: #4CAF50;
      font-weight: bold;
    }
  </style>
</head>
<body>
  <div class="container">
```

```

<div class="card">
  
  <h2>GAMING PRO</h2>
  <p>RTX 4080 gaming laptop</p>
  <p class="price">$999.99</p>
</div>
<div class="card">
  
  <h2>ULTRABOOK</h2>
  <p>Lightweight productivity laptop</p>
  <p class="price">$799.99</p>
</div>
<div class="card">
  
  <h2>CREATOR STUDIO</h2>
  <p>4K video editing laptop</p>
  <p class="price">$1299.99</p>
</div>
<div class="card">
  
  <h2>BUSINESS ELITE</h2>
  <p>Enterprise grade laptop</p>
  <p class="price">$899.99</p>
</div>
</div>
</body>
</html>

```

smartphones.html

```

<!DOCTYPE html>
<html>
<head>
  <title>Smartphones</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 20px;
      background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
    }
    .container {
      display: grid;
      grid-template-columns: 1fr 1fr;
      gap: 20px;
    }
  </style>

```

```
.card {
  background: white;
  padding: 20px;
  border-radius: 15px;
  box-shadow: 0 8px 25px rgba(0,0,0,0.1);
  text-align: center;
}
.card-img {
  width: 200px;
  height: 200px;
  border-radius: 12px;
  margin-bottom: 15px;
}
.price {
  font-size: 20px;
  color: #4CAF50;
  font-weight: bold;
}
</style>
</head>
<body>
  <div class="container">
    <div class="card">
      
      <h2>ULTRA PRO</h2>
      <p>Latest flagship smartphone</p>
      <p class="price">$799.99</p>
    </div>
    <div class="card">
      
      <h2>MIDRANGE X</h2>
      <p>Best value smartphone</p>
      <p class="price">$499.99</p>
    </div>
    <div class="card">
      
      <h2>GAMER PHONE</h2>
      <p>120Hz gaming smartphone</p>
      <p class="price">$649.99</p>
    </div>
    <div class="card">
      
      <h2>FOLD PRO</h2>
      <p>Foldable display smartphone</p>
      <p class="price">$999.99</p>
    </div>
  </div>
</body>
```

```
</html>
```

smartwatches.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Smart Watches</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 20px;
      background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
    }
    .container {
      display: grid;
      grid-template-columns: 1fr 1fr;
      gap: 20px;
    }
    .card {
      background: white;
      padding: 20px;
      border-radius: 15px;
      box-shadow: 0 8px 25px rgba(0,0,0,0.1);
      text-align: center;
    }
    .card-img {
      width: 200px;
      height: 200px;
      border-radius: 12px;
      margin-bottom: 15px;
    }
    .price {
      font-size: 20px;
      color: #4CAF50;
      font-weight: bold;
    }
  </style>
</head>
<body>
  <div class="container">
    <div class="card">
      
      <h2>PREMIUM S9</h2>
```

```

        <p>Advanced health tracking watch</p>
        <p class="price">$199.99</p>
    </div>
    <div class="card">
        
        <h2>FIT TRACKER</h2>
        <p>24/7 fitness monitoring</p>
        <p class="price">$129.99</p>
    </div>
    <div class="card">
        
        <h2>SPORTS PRO</h2>
        <p>GPS running and cycling watch</p>
        <p class="price">$249.99</p>
    </div>
    <div class="card">
        
        <h2>CLASSIC SERIES</h2>
        <p>Traditional smartwatch design</p>
        <p class="price">$179.99</p>
    </div>
</div>
</body>
</html>

```

headphones.html

```

<!DOCTYPE html>
<html>
<head>
    <title>Headphones</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 20px;
            background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
        }
        .container {
            display: grid;
            grid-template-columns: 1fr 1fr;
            gap: 20px;
        }
        .card {
            background: white;
            padding: 20px;
            border-radius: 15px;
            box-shadow: 0 8px 25px rgba(0,0,0,0.1);
        }
    </style>

```

```

        text-align: center;
    }
    .card-img {
        width: 200px;
        height: 200px;
        border-radius: 12px;
        margin-bottom: 15px;
    }
    .price {
        font-size: 20px;
        color: #4CAF50;
        font-weight: bold;
    }
</style>
</head>
<body>
    <div class="container">
        <div class="card">
            
            <h2>PREMIUM OVER-EAR</h2>
            <p>ANC wireless headphones</p>
            <p class="price">$149.99</p>
        </div>
        <div class="card">
            
            <h2>TRUE WIRELESS</h2>
            <p>Compact earbuds with case</p>
            <p class="price">$79.99</p>
        </div>
        <div class="card">
            
            <h2>GAMING HEADSET</h2>
            <p>7.1 surround sound headset</p>
            <p class="price">$99.99</p>
        </div>
        <div class="card">
            
            <h2>STUDIO PRO</h2>
            <p>Professional monitoring headphones</p>
            <p class="price">$199.99</p>
        </div>
    </div>
</body>
</html>

```

Output:

GADGET HUB - Brave

C:/Users/asus/Documents/UID_CSE25223/Week-6/6.2.html

Gmail youtube WhatsApp Mail - Padmanabhu... Free Programming... GitHub - yeyintmint... My Dashboard - IIT... Tutedude: Online C... Free Online Skills Tr... My Programs | WISE... All Bookmarks

GADGET HUB

Categories

LAPTOPS

SMARTPHONES

HEADPHONES

SMART WATCHES



GAMING PRO
RTX 4080 gaming laptop
\$999.99



ULTRABOOK
Lightweight productivity laptop
\$799.99

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GADGET HUB - Brave

C:/Users/asus/Documents/UID_CSE25223/Week-6/6.2.html

Gmail youtube WhatsApp Mail - Padmanabhu... Free Programming... GitHub - yeyintmint... My Dashboard - IIT... Tutedude: Online C... Free Online Skills Tr... My Programs | WISE... All Bookmarks

GADGET HUB

OFF ON GADGETS 🔥

Categories

LAPTOPS

SMARTPHONES

HEADPHONES

SMART WATCHES

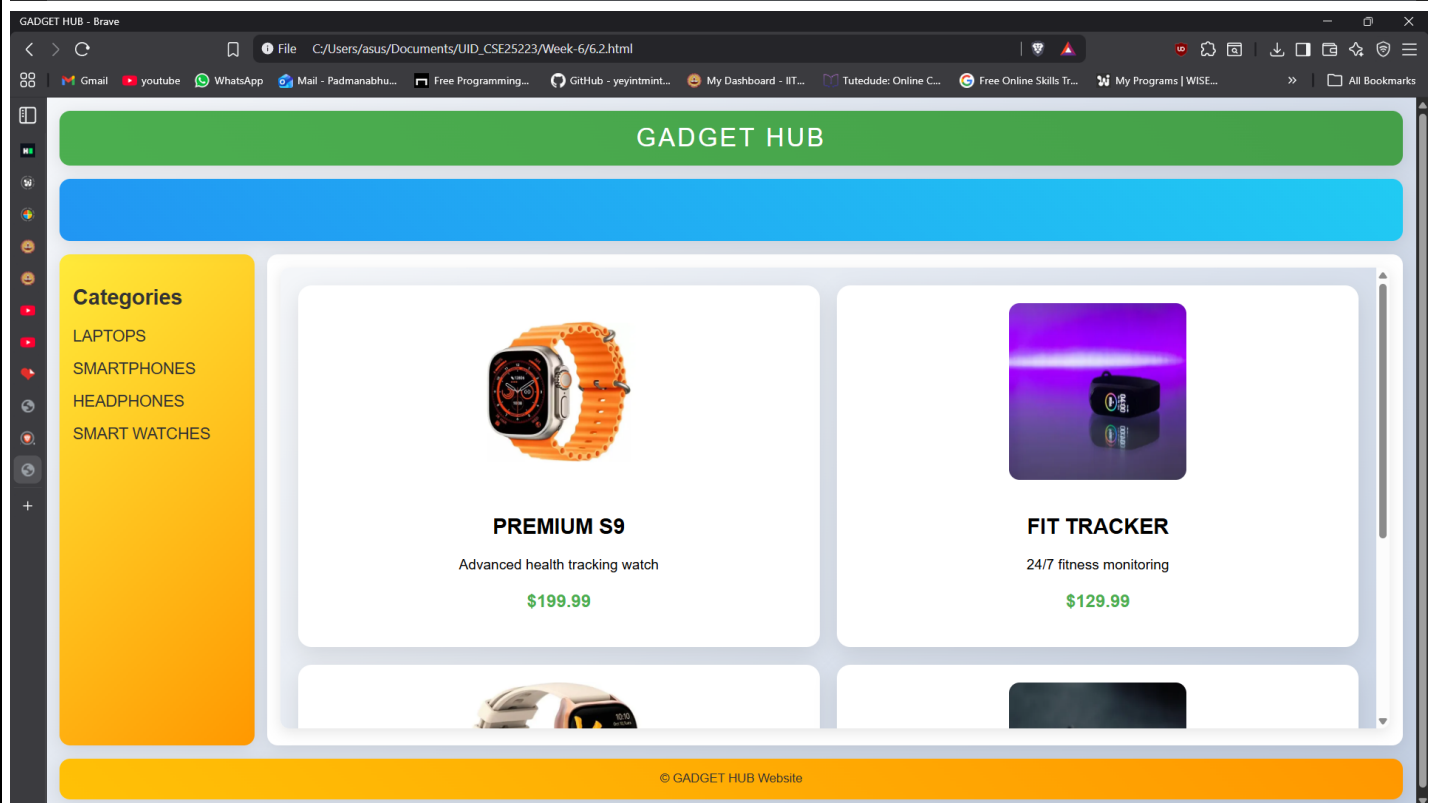
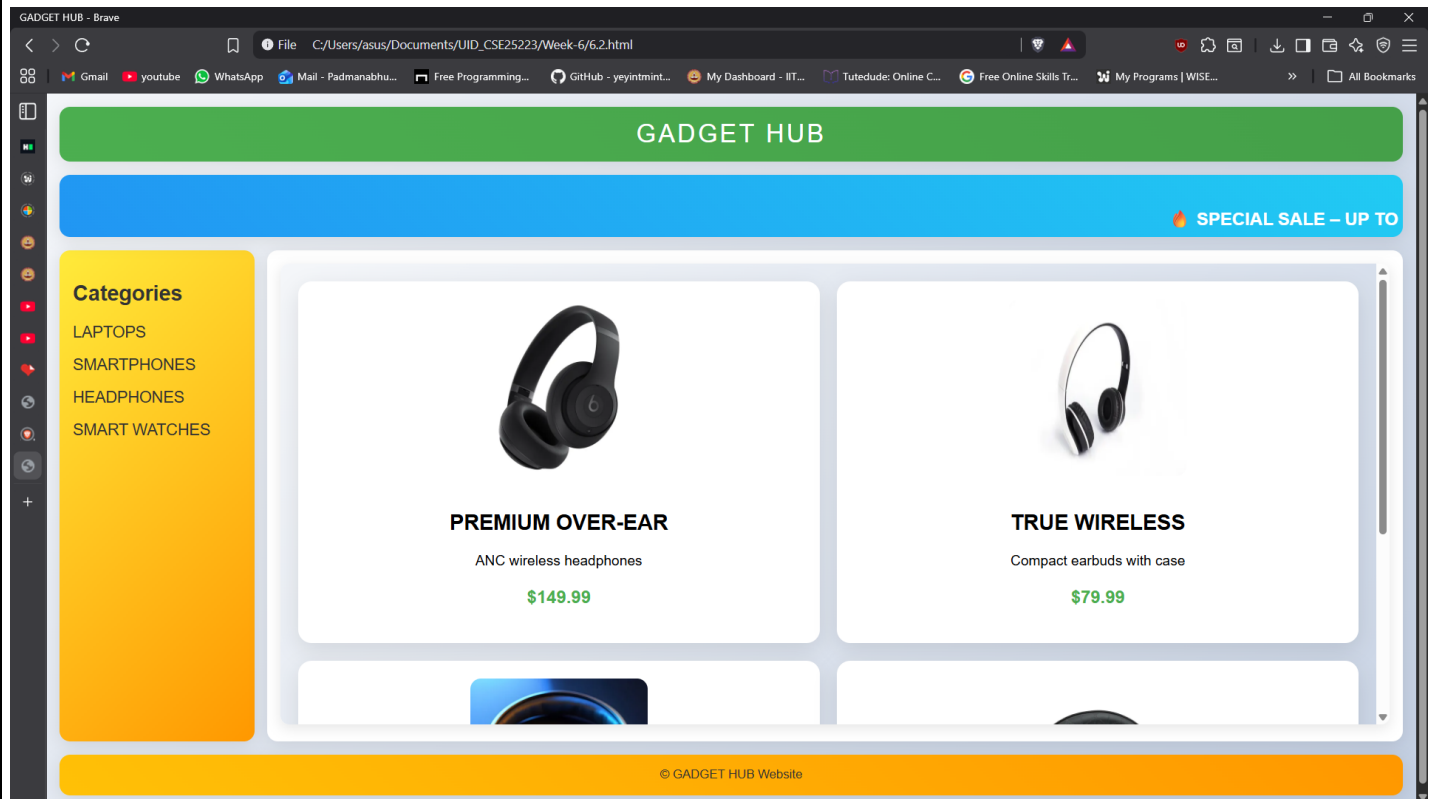


ULTRA PRO
Latest flagship smartphone
\$799.99



MIDRANGE X
Best value smartphone
\$499.99

© GADGET HUB Website



Explanation of the tags:

- **target="contentFrame":** Loads linked pages inside specific iframe by name
- **name="contentFrame":** Names iframe so menu links can target it

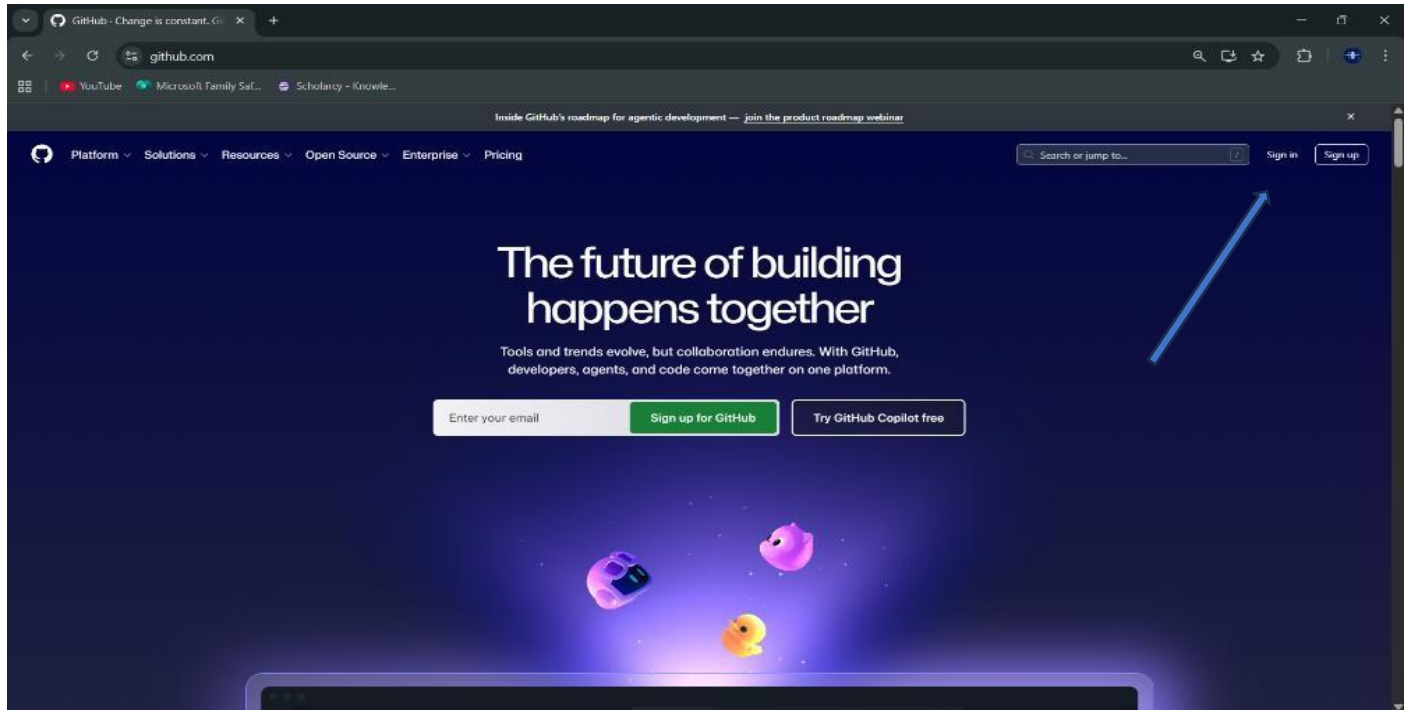
- **grid-template:** CSS Grid shorthand for rows, columns, and areas
- **grid-area: header:** Assigns element to named grid area
- **linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%):** Light theme background gradient
- **@keyframes move:** Creates scrolling sale text animation
- **animation: move 10s linear infinite:** Continuous movement of sale banner text
- **box-shadow: 0 4px 20px rgba(0,0,0,0.1):** Adds soft shadow around sections/cards
- **transition: .3s:** Smooth hover effect on sidebar links
- **display: block:** Makes sidebar links take full line width
- **border-radius: 10px:** Rounds iframe corners
- **grid-template-columns: 1fr 1fr:** Creates 2-column product layout in sub-pages
- **rgba(255,255,255,.95):** Semi-transparent white section background
- **font-weight: bold:** Makes text thicker for emphasis
- **letter-spacing: 3px:** Adds spacing between header letters

WEEK-7

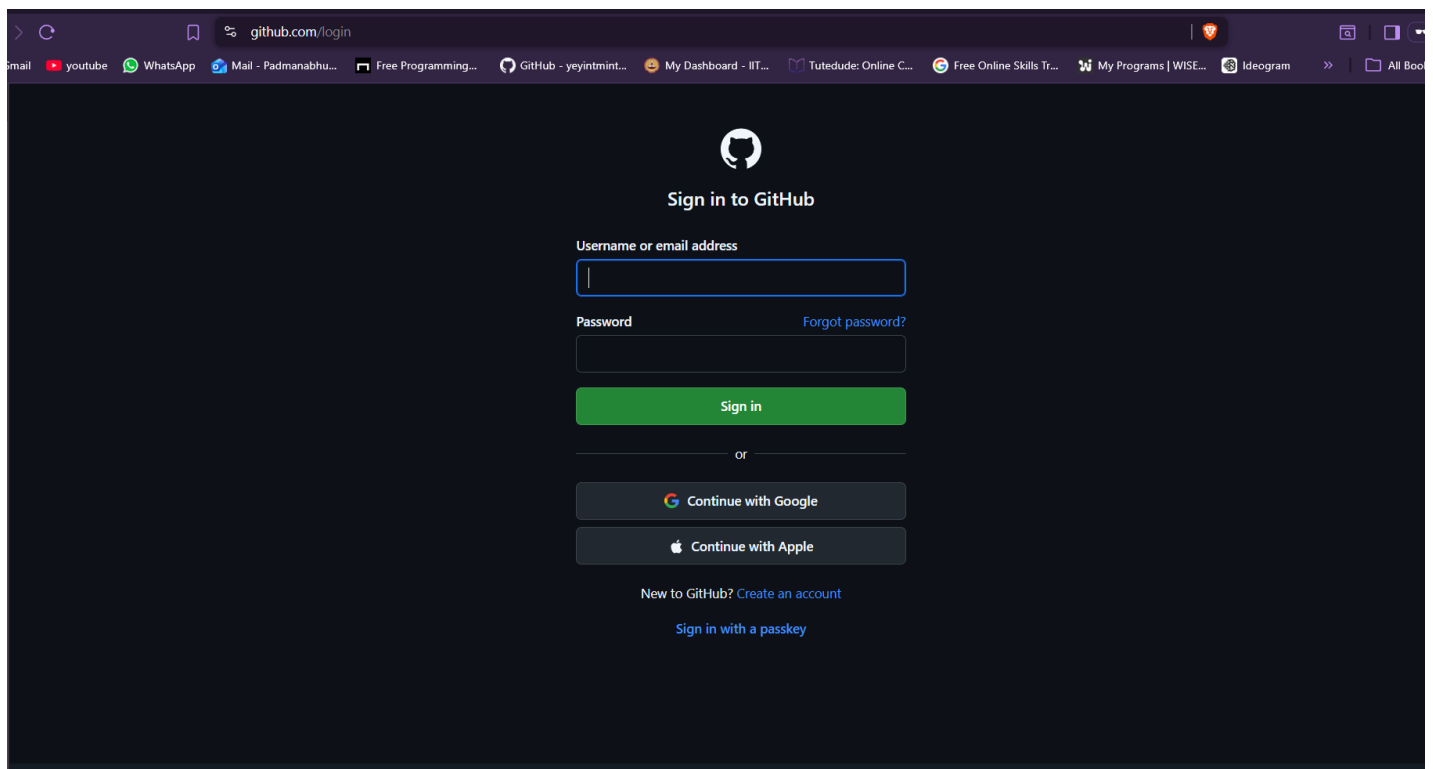
7.Aim: Create a Github Repository and upload all related files of lab manual.

Procedure:

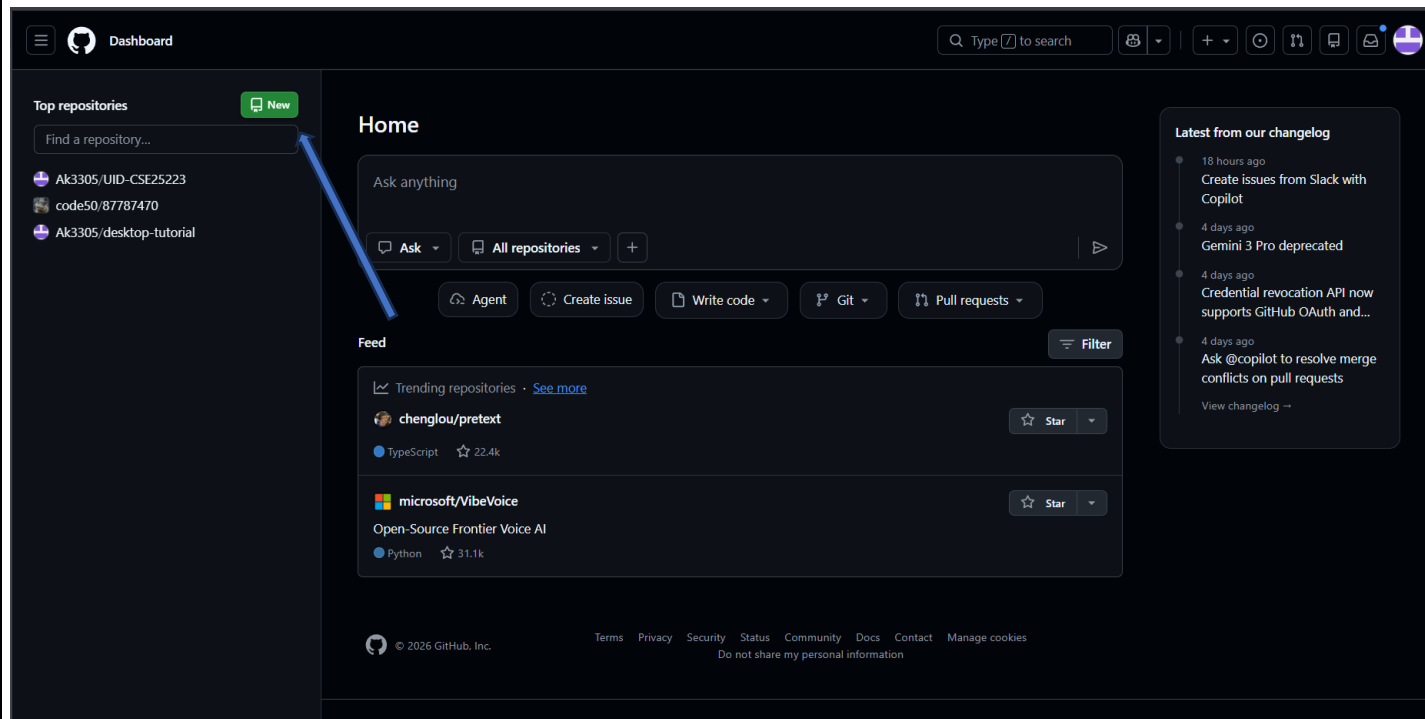
Step-1: Browse to [Github](https://github.com) and create an Account.



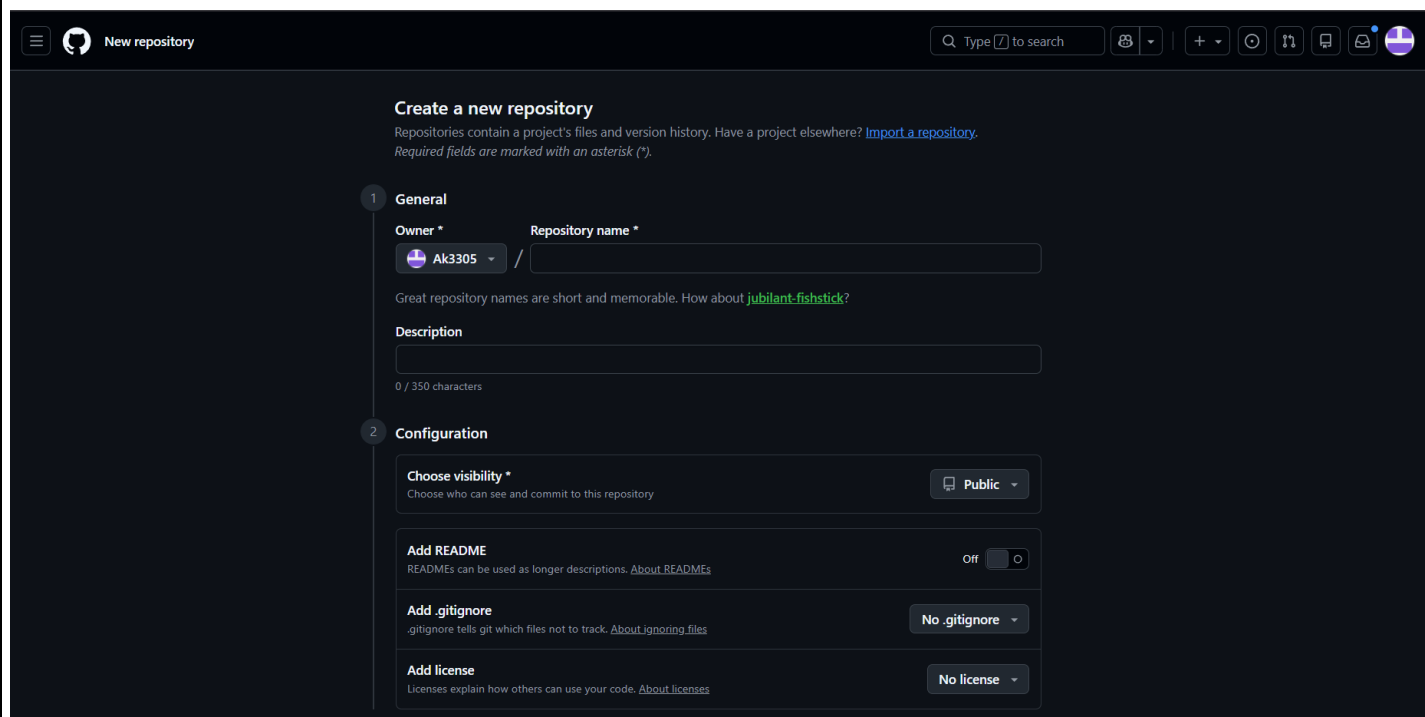
Step-2: Login to Github and Create a Repository



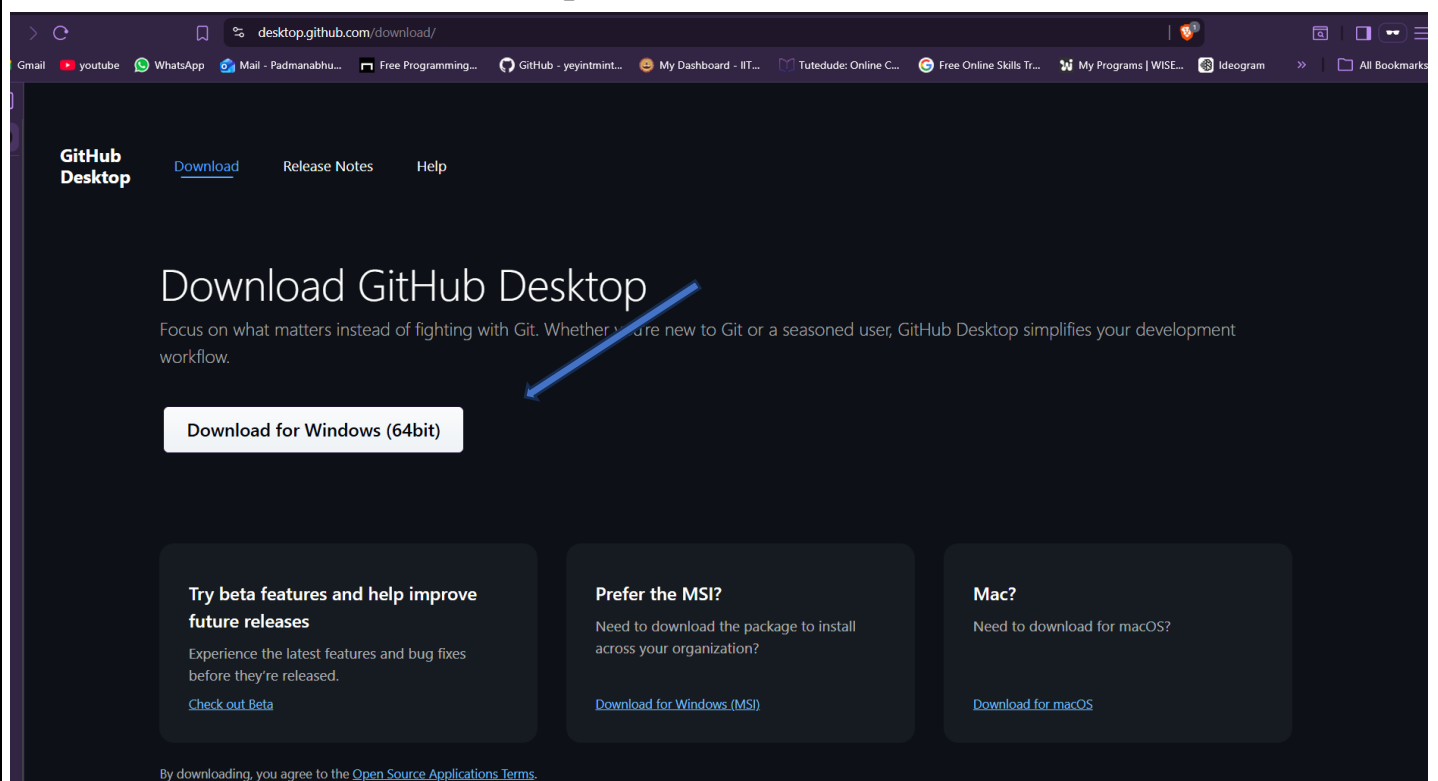
Step-3: Create a repository after login by clicking new.



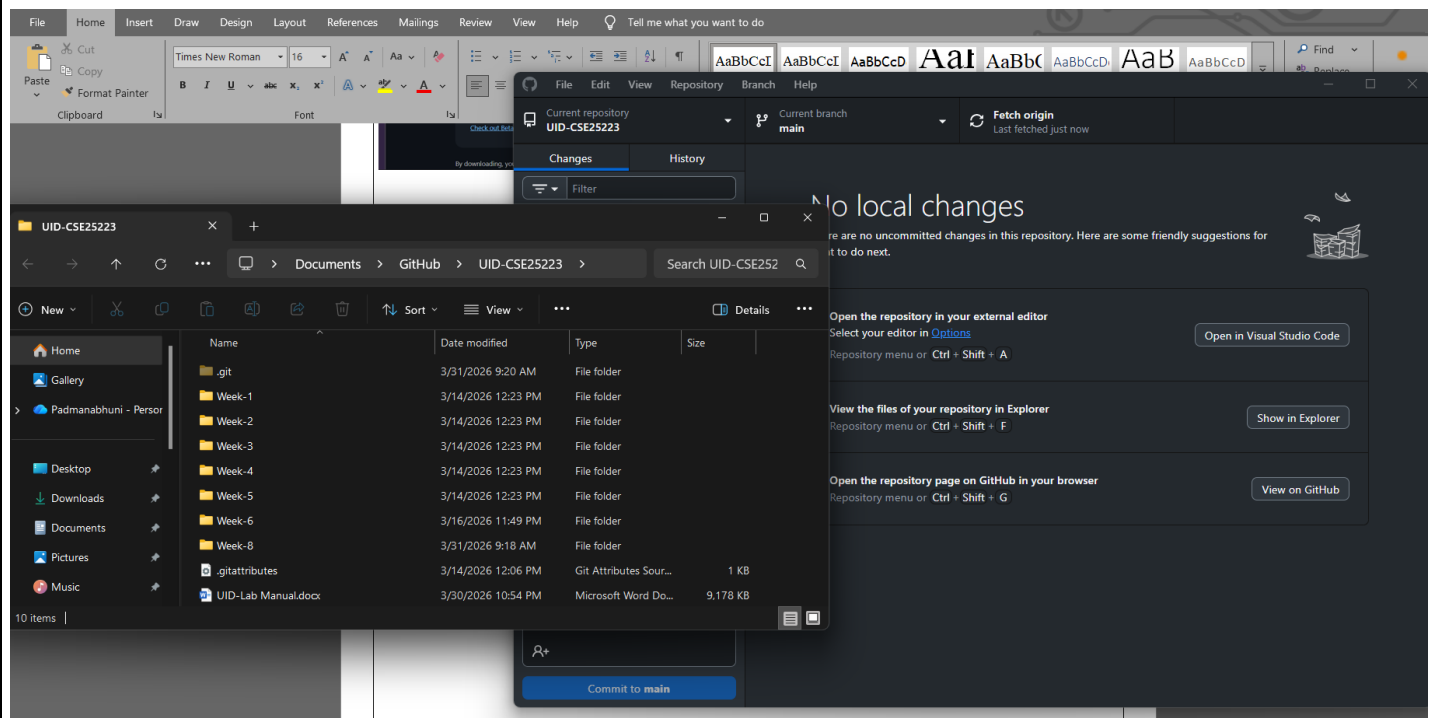
Step-4: Specify the Repository specifications



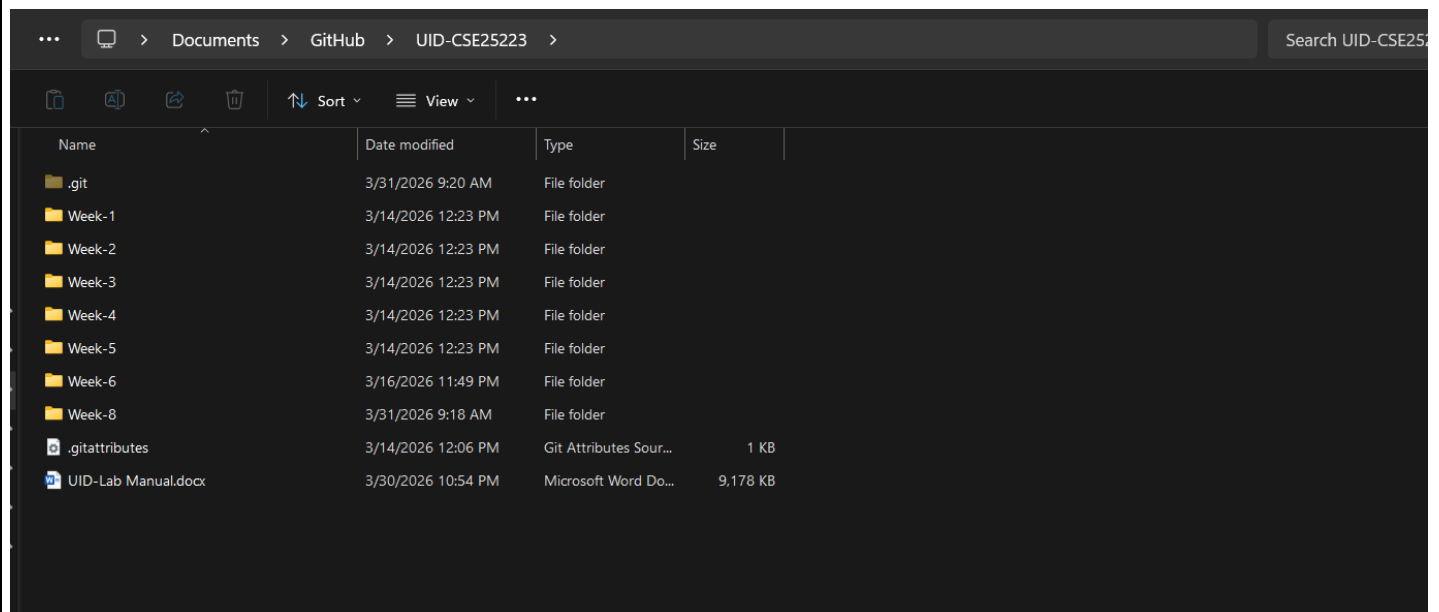
Step-5: Download Github Desktop and Launch it



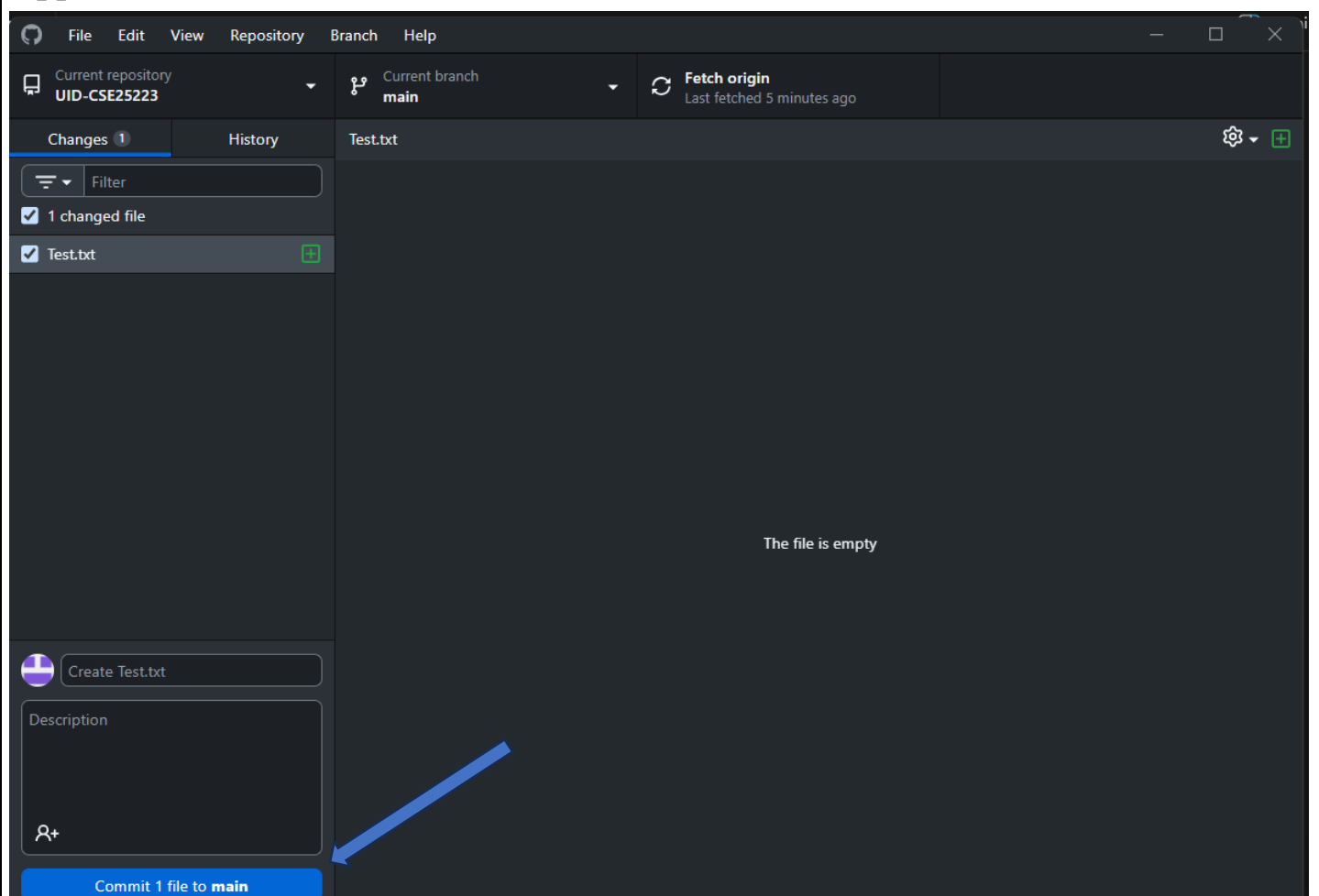
Step-6: In Github Desktop app setup a folder from your explorer.

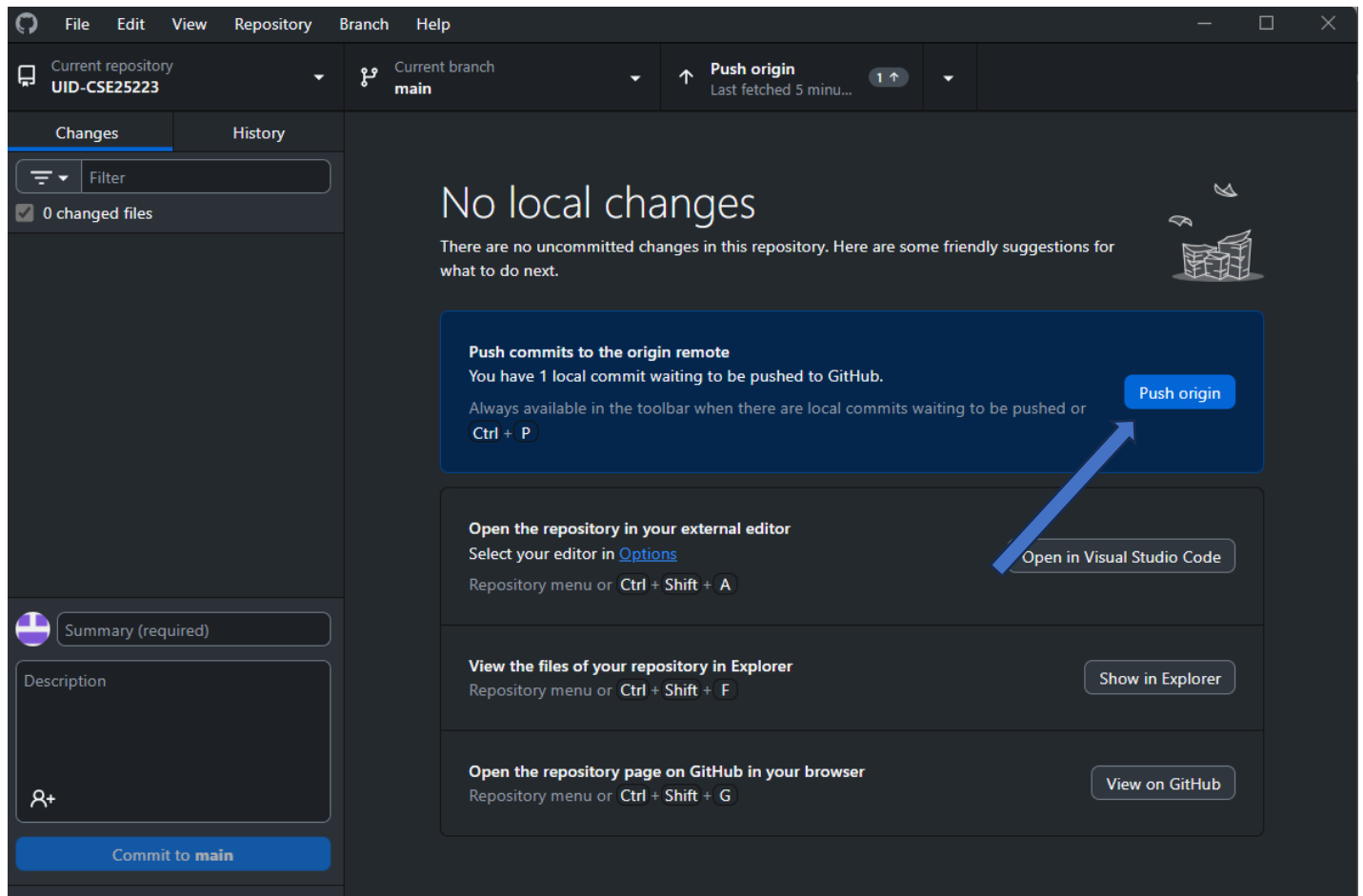


Step-7: Add all Week files to the folder linked to Github.

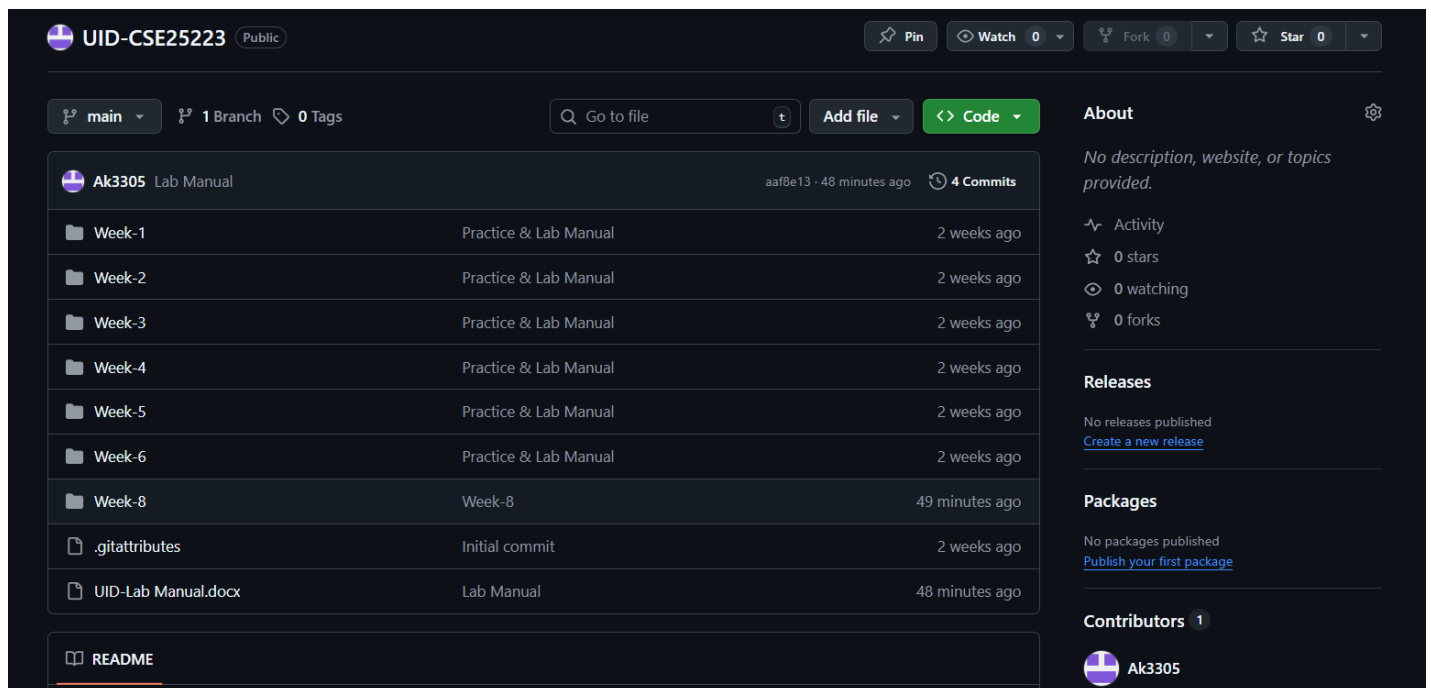


Step-8: Now click on Commit changes and Push the files in the GithubDesktop Applications.





Step-9: Now go to browser and check it. All the new added files in the repository .



Link:- <https://github.com/Ak3305/UID-CSE25223>

WEEK-8

8.A.Aim: Create a webpage that displays a square that continuously rotates 360 degree using CSS Animation.

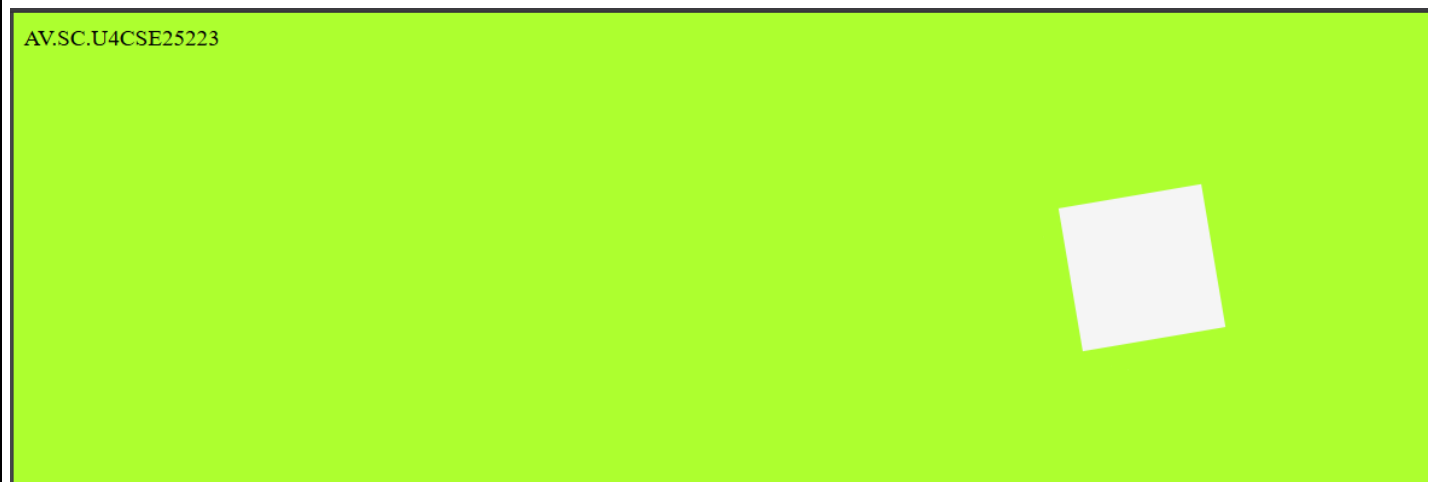
Code:

```
<!DOCTYPE html>
<head>
  AV.SC.U4CSE25223
<style>
div {
  width: 100px;
  height: 100px;
  background: whitesmoke;
  margin: 100px auto;
  animation: mymove 5s infinite linear;
}

@keyframes mymove {
  from {transform: rotate(0deg);}
  to {transform: rotate(360deg);}
}
</style>
</head>

<body style="background-color: greenyellow;">
<div></div>
</body>
```

Output:



Explanation of the tags:

- animation: mymove 5s infinite linear: Runs animation for 5s, repeats forever, linear speed.

- @keyframes mymove: Defines animation steps from start to end
- transform: rotate(0deg): Rotates element 0 degrees (start position)
- transform: rotate(360deg): Rotates element full circle (end position)
- width: 100px; height: 100px: Creates square shape
- margin: 100px auto: Centers square horizontally, adds top spacing
- background: whitesmoke: Light gray square color

8.B.Aim: Create a ball that moves up and down continuously using CSS Animation.

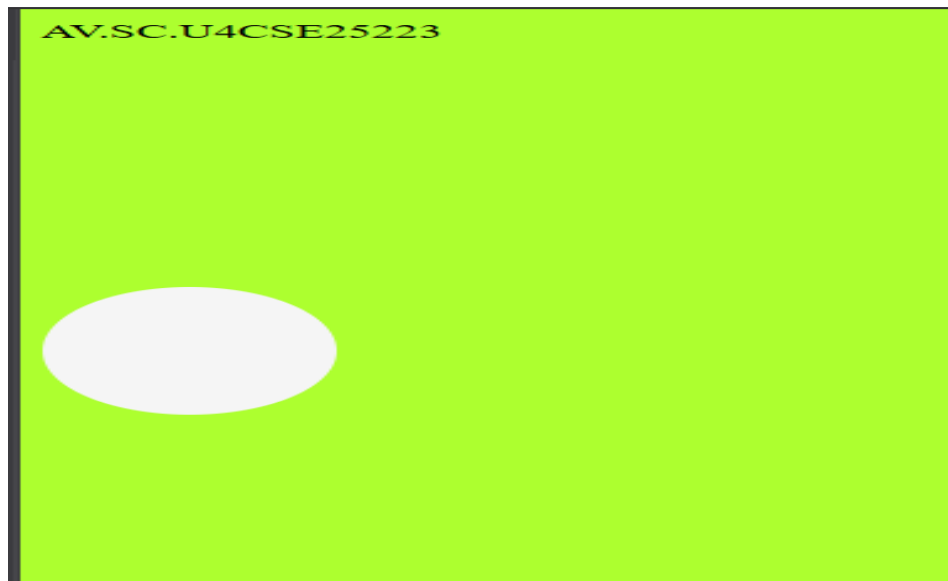
Code:

```
<!DOCTYPE html>
<head>
  AV.SC.U4CSE25223
<style>
div {
  width: 100px;
  height: 100px;
  background: whitesmoke;
  border-radius: 50%;
  position: relative;
  animation: mymove 2s infinite alternate;
}

@keyframes mymove {
  from {top: 0px;}
  to {top: 200px;}
}
</style>
</head>

<body style="background-color: greenyellow;">
<div></div>
</body>
```

Output:



Explanation of the tags:

- border-radius: 50%: Makes square into perfect circle
- position: relative: Allows top positioning from original spot
- animation: mymove 2s infinite alternate: Bounces up/down every 2s, reverses direction
- top: 0px: Starting position at top
- top: 200px: Ending position 200px down
- @keyframes mymove: Defines bounce path from top to bottom
- width/height: Creates circular ball shape
- infinite alternate: Keeps bouncing back and forth

8.C.Aim: Create a circle that moves horizontally and rotates at the same time using CSS Animation.

Code:

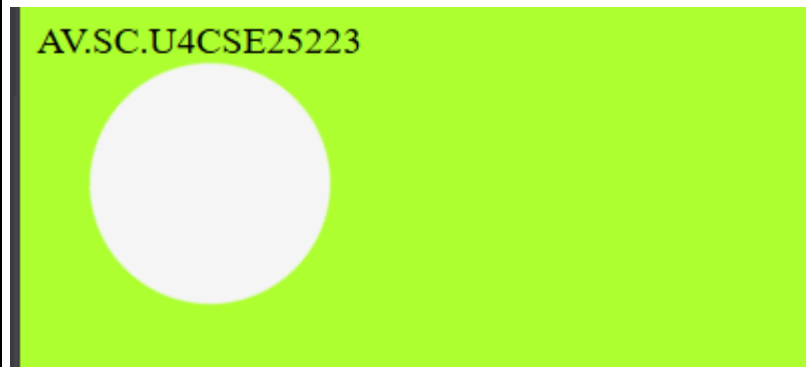
```
<!DOCTYPE html>
<head>
  AV.SC.U4CSE25223
<style>
div {
  width: 100px;
  height: 100px;
  background: whitesmoke;
  border-radius: 50%;
  position: relative;
  animation: mymove 5s infinite linear;
}
```

```

@keyframes mymove {
  from {
    left: 0px;
    transform: rotate(0deg);
  }
  to {
    left: 200px;
    transform: rotate(360deg);
  }
}
</style>
</head>
<body style="background-color: greenyellow;">
<div></div>
</body>

```

Output:



Explanation of the tags:

- border-radius: 50%: Makes square into perfect circle
- position: relative: Allows top positioning from original spot
- animation: mymove 2s infinite alternate: Bounces up/down every 2s, reverses direction
- top: 0px: Starting position at top
- top: 200px: Ending position 200px down
- @keyframes mymove: Defines bounce path from top to bottom
- width/height: Creates circular ball shape
- infinite alternate: Keeps bouncing back and forth

WEEK-9

9.Aim: Create a webpage using HTML, CSS, JS based on the give conditions:

- a) Create a button for each program, based on the button, clicked the corresponding program has to be executed and should display the output.
- b)The logics for each button includes finding an even or odd number, prime or not, factorial of a number, Armstrong or not.
- c)Use appropriate design to display the output.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Logic Programs</title>
  <style>
    body {
      margin: 0;
      font-family: Arial, sans-serif;
      background: #e6ecf5;
      display: flex;
      justify-content: center;
      align-items: center;
      min-height: 100vh;
    }
    .container {
      background: #ffffff;
      border-radius: 12px;
      box-shadow: 0 4px 20px rgba(0,0,0,0.2);
      padding: 40px;
      width: 380px;
      text-align: center;
    }
    h2 {
      margin-bottom: 20px;
      color: #2c3e50;
    }
    input {
      width: 100%;
      padding: 10px;
      font-size: 16px;
      border: 2px solid #a7b6c7;
      border-radius: 8px;
      margin-bottom: 20px;
      box-sizing: border-box;
      text-align: center;
    }
```

```

        background: #f8fbff;
        color: #333;
    }
    input:focus {
        outline: none;
        border-color: #5b8def;
    }
    .buttons {
        display: grid;
        grid-template-columns: 1fr 1fr;
        gap: 12px;
        margin-bottom: 24px;
    }
    button {
        padding: 12px;
        font-size: 15px;
        border: none;
        border-radius: 8px;
        cursor: pointer;
        color: white;
        font-weight: bold;
        background: #5b8def;
        transition: opacity 0.2s;
    }
    button:hover { opacity: 0.85; }
    .output {
        background: #eef3f8;

        padding: 20px;
        border-radius: 8px;
        min-height: 60px;
        color: #2f3b45;
        font-size: 16px;
        border: 1px solid #c9d6e2;
    }
</style>
</head>
<body>

    <div class="container">
        <h2>Logic Programs</h2>
        <input type="number" id="numInput" placeholder="Enter a number" />
        <div class="buttons">
            <button onclick="checkEvenOdd()">Even / Odd</button>

```

```

<button onclick="checkPrime()">Prime / Not</button>

<button onclick="calcFactorial()">Factorial</button>
<button onclick="checkArmstrong()">Armstrong</button>
</div>
<div class="output" id="output">Result will appear here</div>
</div>
<script>
function getInput() {
  const val = document.getElementById('numInput').value.trim();
  if (val === '' || isNaN(val)) {
    document.getElementById('output').textContent = 'Please enter a valid
number.';
    return null;
  }
  return parseInt(val);
}
function setOutput(text) {
  document.getElementById('output').textContent = text;
}
function checkEvenOdd() {
  const n = getInput();
  if (n === null) return;
  setOutput(n + ' is ' + (n % 2 === 0 ? 'Even' : 'Odd'));
}
function checkPrime() {
  const n = getInput();
  if (n === null) return;
  if (n < 2) { setOutput(n + ' is Not Prime'); return; }
  for (let i = 2; i <= Math.sqrt(n); i++) {
    if (n % i === 0) { setOutput(n + ' is Not Prime'); return; }
  }
  setOutput(n + ' is Prime');
}
function calcFactorial() {
  const n = getInput();
  if (n === null) return;
  if (n < 0) { setOutput('Factorial not defined for negative numbers'); return; }
  let result = 1;
  for (let i = 2; i <= n; i++) result *= i;
  setOutput(n + ' ! = ' + result);
}
function checkArmstrong() {
  const n = getInput();
  if (n === null) return;

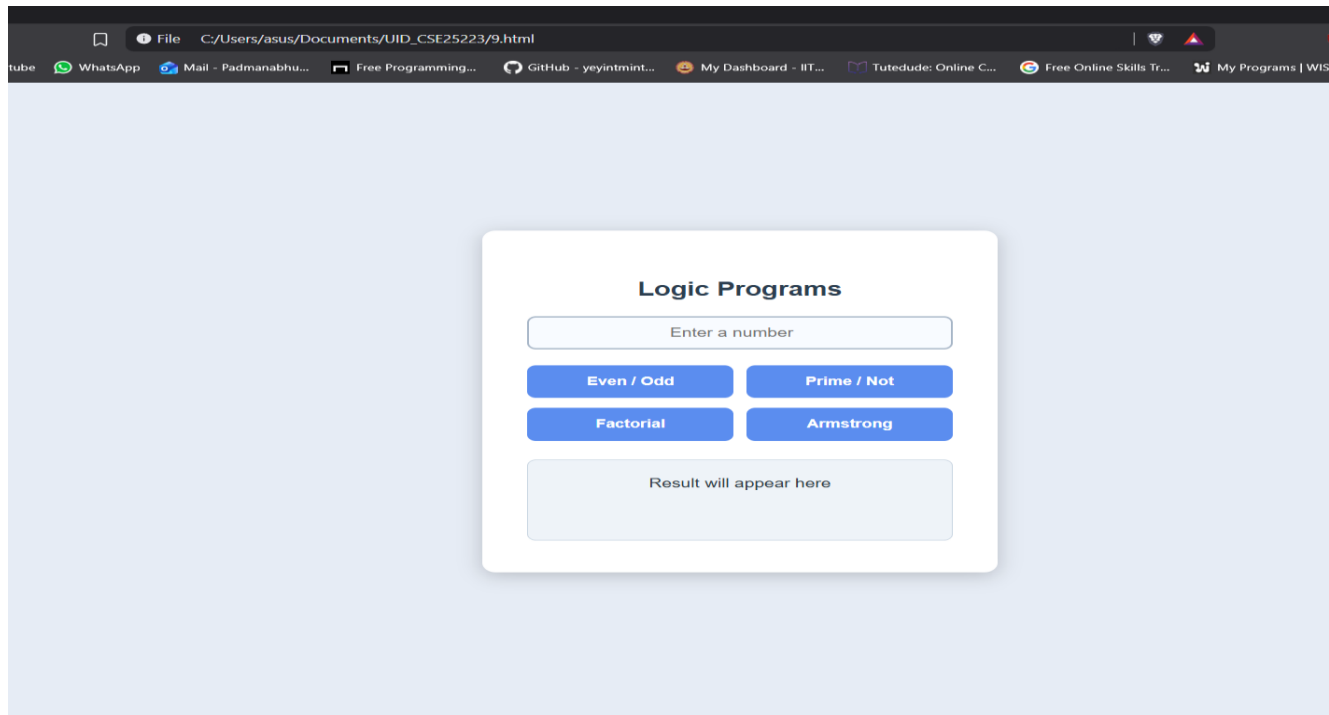
```

```

        const digits = String(n).split('');
        const power  = digits.length;
        const sum     = digits.reduce((acc, d) => acc + Math.pow(Number(d), power), 0);
        setOutput(n + (sum === n ? ' is an Armstrong number' : ' is Not an Armstrong
number'));
    }
</script>
</body>
</html>

```

Output:



Explanation of the tags:

- `<html>` – Top-level container for the entire HTML page.
- `<head>` – Holds metadata, title, styles, and scripts.
- `<title>` – Sets the page title shown in the browser tab.
- `<body>` – Contains all visible content of the page.
- `<div class="container">` – Groups the main UI into a styled card.
- `<h2>` – Second-level heading for “Logic Programs”.
- `<input type="number">` – Text box that accepts only numeric input.
- `id="numInput"` – Unique name used in JavaScript to access the input.
- `onclick="checkEvenOdd()"` – Runs the checkEvenOdd function when the button is clicked.

WEEK-10

10.Aim: Write JavaScript conditional statement to find the sign of product of three numbers using functions .Display an alert box with specified sign. Write a java script conditional statements to sort 3 numbers using functions display an alert box to show the result.write a java script conditional statement to find largest of find largest of five numbers using function and onclick event .display an alert box to show the result .write a java script for loop that will iterate from 0 to 15. For each iteration ,it will check if the current number odd or even , and display a message on screen .write a java script program to read 6 subjects marks from the student then computes the average marks of students then, this average is used to determine the corresponding grade.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Logic Programs</title>
  <style>
    body {
      font-family: Arial;
      display: flex;
      height: 100vh;
      background: #edf2f7;
    }
    .left {
      width: 30%;
      background: #dbeafe;
      padding: 20px;
      border-right: 3px solid #93c5fd;
    }
    .right {
      width: 70%;
      background: #f8fafc;
      padding: 20px;
    }
    input {
      width: 95%;
      padding: 10px;
      margin-bottom: 10px;
      background: #ffffff;
      border: 1px solid #cbd5e1;
      color: #1f2937;
    }
    button {
      width: 100%;
```

```

        padding: 10px;
        margin-bottom: 10px;
        background: #2563eb;
        border: none;
        cursor: pointer;
        color: white;
    }
    #output {
        background: #fff6ff;
        padding: 20px;
        min-height: 200px;
        white-space: pre-line;
        color: #111827;
    }
</style>
</head>
<body>
<div class="left">
    <h3>Enter Input</h3>
    <input type="text" id="numbers" placeholder="Click a button to see input format">
    <button onclick="signProduct()">Sign of Product</button>
    <button onclick="sortNumbers()">Sort Numbers</button>
    <button onclick="largestNumber()">Largest Number</button>
    <button onclick="oddEvenLoop()">Odd / Even 0-15</button>
    <button onclick="studentGrade()">Student Grade</button>
</div>
<div class="right">
    <h3>Output</h3>
    <div id="output">Result will appear here</div>
</div>
<script>
function getNumbers() {
    let input = document.getElementById("numbers").value;
    return input.split(",").map(Number);
}
function signProduct() {
    let input = document.getElementById("numbers").value;
    if (input === "") {
        document.getElementById("output").textContent =
            "Enter 3 numbers like: 0,-1,4";
        return;
    }
    let nums = getNumbers();
    let negative = 0;
    for (let n of nums) {
        if (n < 0) negative++;
    }
    let result = (negative % 2 == 0) ? "+1" : "-1";

```



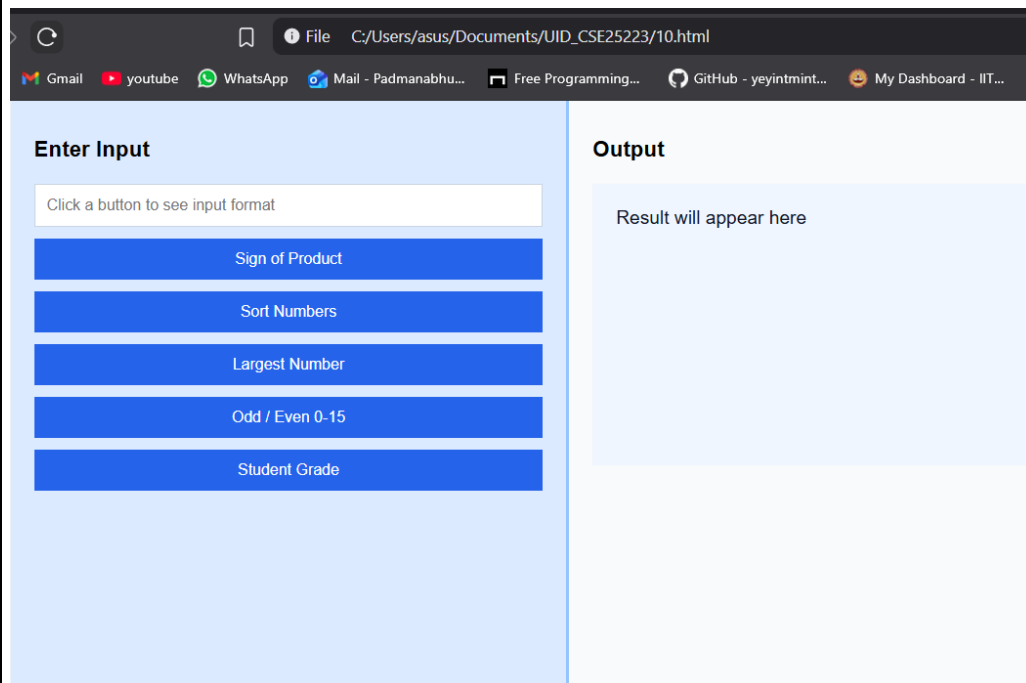
```
    alert(result);
    document.getElementById("output").textContent = "Sign: " + result;
}
function sortNumbers() {
    let input = document.getElementById("numbers").value;
    if (input === "") {
        document.getElementById("output").textContent =
            "Enter 3 numbers like: 0,-1,4";
        return;
    }
    let nums = getNumbers();
    nums.sort((a, b) => b - a);
    alert(nums.join(", "));
    document.getElementById("output").textContent =
        "Sorted: " + nums.join(", ");
}
function largestNumber() {
    let input = document.getElementById("numbers").value;
    if (input === "") {
        document.getElementById("output").textContent =
            "Enter 5 numbers like: -5,-2,-6,0,-1";
        return;
    }
    let nums = getNumbers();
    let max = nums[0];
    for (let n of nums) {
        if (n > max) max = n;
    }
    alert("Largest: " + max);
    document.getElementById("output").textContent =
        "Largest: " + max;
}
function oddEvenLoop() {
    let input = document.getElementById("numbers").value;
    if (input === "") {
        document.getElementById("output").textContent =
            "Enter a number like: 15 (to check from 0 to that number)";
        return;
    }
    let n = Number(input);
    let result = "";
    for (let i = 0; i <= n; i++) {
        if (i % 2 == 0)
            result += i + " is Even\n";
        else
            result += i + " is Odd\n";
    }
    document.getElementById("output").textContent = result;
}
```

```

}
function studentGrade() {
    let input = document.getElementById("numbers").value;
    if (input === "") {
        document.getElementById("output").textContent =
            "Enter 6 subject marks like: 70,80,90,60,75,85";
        return;
    }
    let nums = getNumbers();
    let sum = 0;
    for (let n of nums) sum += n;
    let avg = sum / nums.length;
    let grade;
    if (avg >= 90) grade = "A";
    else if (avg >= 75) grade = "B";
    else if (avg >= 60) grade = "C";
    else grade = "D";
    alert("Average: " + avg + " Grade: " + grade);
    document.getElementById("output").textContent =
        "Average: " + avg + " | Grade: " + grade;
}
</script>
</body>
</html>

```

Output:



Explanation of the tags:

- `getNumbers()` – Gets the text from the input box, splits it by commas, and converts each value into a number.

- `signProduct()` – Checks how many negative numbers are entered and decides whether the product sign is +1 or -1.
- `sortNumbers()` – Takes the entered numbers and sorts them in descending order using `sort((a, b) => b - a)`.
- `largestNumber()` – Finds the biggest value by comparing each number one by one in a loop.
- `oddEvenLoop()` – Prints whether every number from 0 to the entered value is even or odd.
- `studentGrade()` – Calculates the average of all marks and assigns a grade based on that average.
- `document.getElementById("numbers")` – Selects the input element with id numbers so JavaScript can read its value.
- `input.split(",")` – Breaks a comma-separated string into separate parts.
- `map(Number)` – Converts each split value from text into a numeric value.
- `white-space: pre-line;` – Keeps line breaks in the output box so each even/odd result appears on a new line.

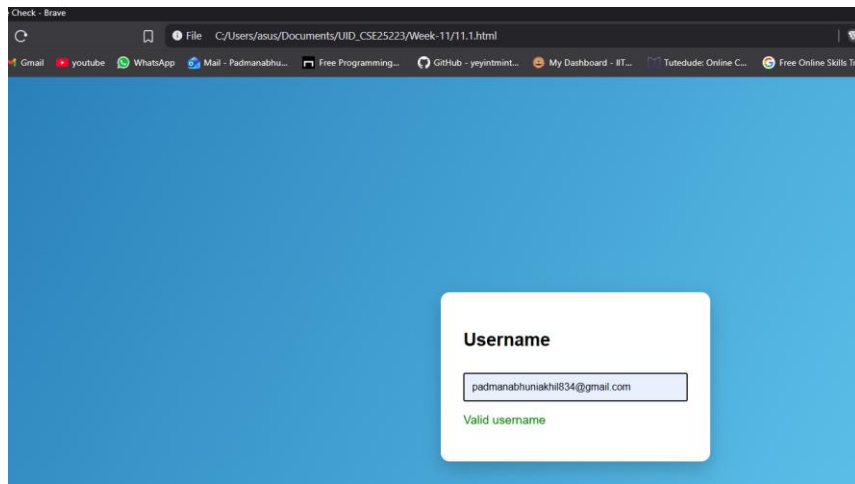
WEEK-11

11.A.Aim: A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters.

Code:

```
<html>
<head>
<title>Username Check</title>
<style>
body {
  margin:0;
  font-family: Arial;
  background: linear-gradient(120deg,#2980b9,#6dd5fa);
  display:flex;
  justify-content:center;
  align-items:center;
  height:100vh;
}
.card {
  background:white;
  padding:30px;
  border-radius:12px;
  width:300px;
  box-shadow:0 10px 25px rgba(0,0,0,0.2);
}
input { width:100%; padding:10px; margin-top:10px; }
</style>
</head>
<body>
<div class="card">
<h2>Username</h2>
<input type="text" id="username" placeholder="Enter username"
onkeyup="checkUsername()">
<p id="msg"></p>
</div>
<script>
function checkUsername() {
  let val=document.getElementById("username").value;
  let msg=document.getElementById("msg");
  if(val.length<5){ msg.innerText="Minimum 5 characters required";
msg.style.color="red"; }
  else{ msg.innerText="Valid username"; msg.style.color="green"; }
}
</script>
</body>
</html>
```

Output:



Explanation of the tags:

- function : used to define a block of code that performs username validation
- getElementById() : used to access username input and message element
- .value : used to get value from input box
- length : used to check number of characters entered
- if condition : used to check if username is less than 5 characters
- innerText : used to display message on screen
- style.color : used to change text color based on validation

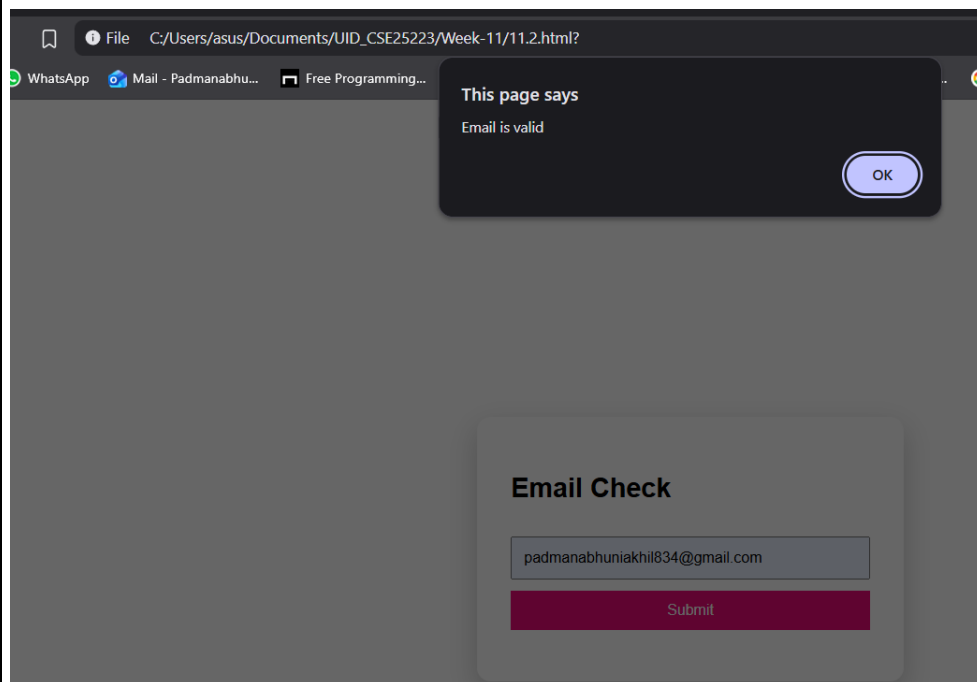
B.Aim: A registration form should validate email format before submission. If invalid, the form should not be submitted.

Code:

```
<html>
<head>
<title>Email Validation</title>
<style>
body{margin:0;font-family:Arial;background:linear
gradient(120deg,#ff6a00,#ee0979);display:flex;justify-content:center;align-
items:center;height:100vh;}
.card{background:white;padding:30px;border-radius:12px;width:320px;box-shadow:0
10px 25px rgba(0,0,0,0.2);}
input,button{width:100%;padding:10px;margin-top:10px;}
button{background:#ee0979;color:white;border:none;cursor:pointer;}
</style>
</head>
<body>
<div class="card">
<h2>Email Check</h2>
```

```
<form onsubmit="return validateEmail()">
<input type="text" id="email" placeholder="Enter Email">
<button type="submit">Submit</button>
</form>
</div>
<script>
function validateEmail(){
    let email=document.getElementById("email").value;
    let pattern=/^[^ ]+@[^ ]+\.[a-z]{2,3}$/;
    if(pattern.test(email)){
        alert("Email is valid");
        return true;
    }else{
        alert("Invalid Email");
        return false;
    }
}
</script>
</body>
</html>
```

Output:



Explanation of the tags:

- .value : gets the email value entered by user
- pattern : regular expression used to check valid email format
- test() : checks if email matches pattern
- if condition : checks whether email is valid or not

- alert() : used to display popup message for valid or invalid email
- return false : prevents form submission if invalid
- return true : allows form submission if valid

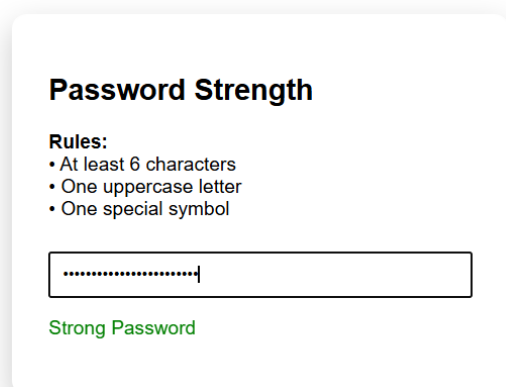
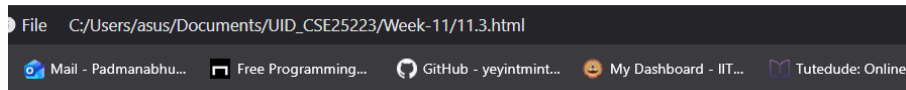
C.Aim: A system should check password strength while typing and display whether it is weak or strong.

Code:

```
<html>
<head>
<title>Password Strength</title>
<style>
body{margin:0;font-family:Arial;background:linear
gradient(120deg,#00b09b,#96c93d);display:flex;justify-content:center;align-
items:center;height:100vh;}
.card{background:white;padding:30px;border-radius:12px;width:350px;box-shadow:0
10px 25px rgba(0,0,0,0.2);}
input{width:100%;padding:10px;margin-top:10px;}
</style>
</head>
<body>
<div class="card">
<h2>Password Strength</h2>
<p><b>Rules:</b><br>• At least 6 characters<br>• One uppercase letter<br>• One
special symbol</p>
<input type="password" id="pass" onkeyup="checkStrength()">
<p id="msg"></p>
</div>
<script>
function checkStrength(){
  let pass=document.getElementById("pass").value;
  let msg=document.getElementById("msg");
  let strong=/^(?=.*[A-Z])(?=.*[!@#$%^&*]).{6,}$/;
  if(strong.test(pass)){
    msg.innerText="Strong Password";
    msg.style.color="green";
  }else{
    msg.innerText="Weak Password";
    msg.style.color="red";
  }
}
</script>
</body>
```

```
</html>
```

Output:



Explanation of the tags:

- pattern : regular expression used to check password strength
- (?.*[A-Z]) : ensures at least one uppercase letter
- (?.*[!@#\$\$%^&*) : ensures at least one special symbol
- .{6,} : ensures minimum 6 characters
- test() : checks if password matches pattern
- if condition : checks strong or weak password
- innerText : displays result message
- style.color : changes text color based on strength

D.Aim: A form should validate a mobile number when the user leaves the input field. The number must be exactly 10 digits.

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Mobile Validation</title>
<style>
body{margin:0;font-family:Arial;background:linear
```

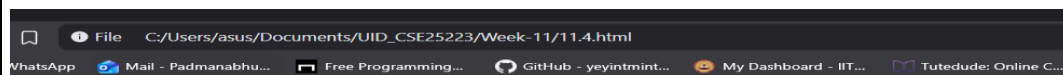


```

gradient(120deg,#fc466b,#3f5efb));display:flex;justify-content:center;align-
items:center;height:100vh;}
.card{background:white;padding:30px;border-radius:12px;width:320px;box-shadow:0
10px 25px rgba(0,0,0,0.2);}
input{width:100%;padding:10px;margin-top:10px;}
</style>
</head>
<body>
<div class="card">
<h2>Mobile & Password</h2>
<input type="text" id="mobile" placeholder="Enter Mobile"
onblur="validateMobile()">
<input type="password" id="pass" placeholder="Enter Password"
onkeyup="checkPass()">
<p id="msg"></p>
</div>
<script>
function validateMobile(){
  let num=document.getElementById("mobile").value;
  let msg=document.getElementById("msg");
  let pattern=/^[0-9]{10}$/; if(pattern.test(num)) msg.innerText="Valid mobile";
  else msg.innerText="Invalid mobile";
}
function checkPass(){
  let pass=document.getElementById("pass").value;
  let strong=/^(?=.*[A-Z])(?=.*[!@#$%^&*]).{6,}$/;
  let msg=document.getElementById("msg");
  if(strong.test(pass)) msg.innerText="Strong Password";
  else msg.innerText="Weak Password";
}
</script>
</body>
</html>

```

Output:



Mobile & Password

Explanation of the tags:

- onblur : triggers validation when user leaves mobile input field
- onkeyup : triggers password validation while typing
- function : defines validation logic
- getElementById() : used to access input fields and message
- .value : gets input values
- pattern : regular expression for mobile validation
- `^[0-9]{10}$` : ensures exactly 10 digits
- test() : checks if mobile number is valid
- if condition : checks valid or invalid mobile password
- pattern : same as strong password rule
- innerText : displays result message

E.Aim: Design an HTML page, such that the registration form must validate:

1. Name (not empty)
2. Email (valid format)
3. Password (minimum 6 characters)

Code:

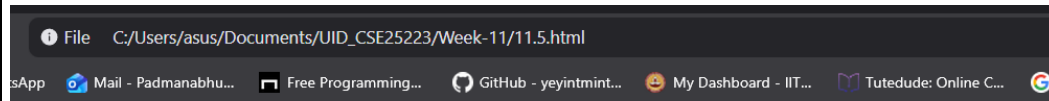
```
<html>
<head>
<title>Register</title>
<style>
body{margin:0;font-family:Arial;background:linear
gradient(120deg,#141e30,#243b55);display:flex;justify-content:center;align-
items:center;height:100vh;color:white;}
.card{background:white;color:black;padding:30px;border-radius:12px;width:350px;box-shadow:0
10px 25px rgba(0,0,0,0.3);}
input,button{width:100%;padding:10px;margin-top:10px;}
button{background:#243b55;color:white;border:none;cursor:pointer;}
</style>
</head>
<body>
<div class="card">
<h2>Registration Form</h2>
<form onsubmit="return validateForm()">
<input type="text" id="name" placeholder="Name">
```

```

<input type="text" id="email" placeholder="Email">
<input type="password" id="pass" placeholder="Password">
<button type="submit">Register</button>
<p id="msg"></p>
</form>
</div>
<script>
function validateForm(){
  let name=document.getElementById("name").value;
  let email=document.getElementById("email").value;
  let pass=document.getElementById("pass").value;
  let msg=document.getElementById("msg");
  let emailPattern=/^[^ ]+@[^ ]+\.[a-z]{2,3}$/;
  let strong=/^(?=.*[A-Z])(?=.*[!@#$%^&*]).{6,}$/;
  if(name===""){ msg.innerText="Name required"; return false; }
  if(!emailPattern.test(email)){ msg.innerText="Invalid email"; return false; }
  if(!strong.test(pass)){ msg.innerText="Weak password"; return false; }
  alert("Registration Successful");
  return true;
}
</script>
</body>
</html>

```

Output:



Registration Form

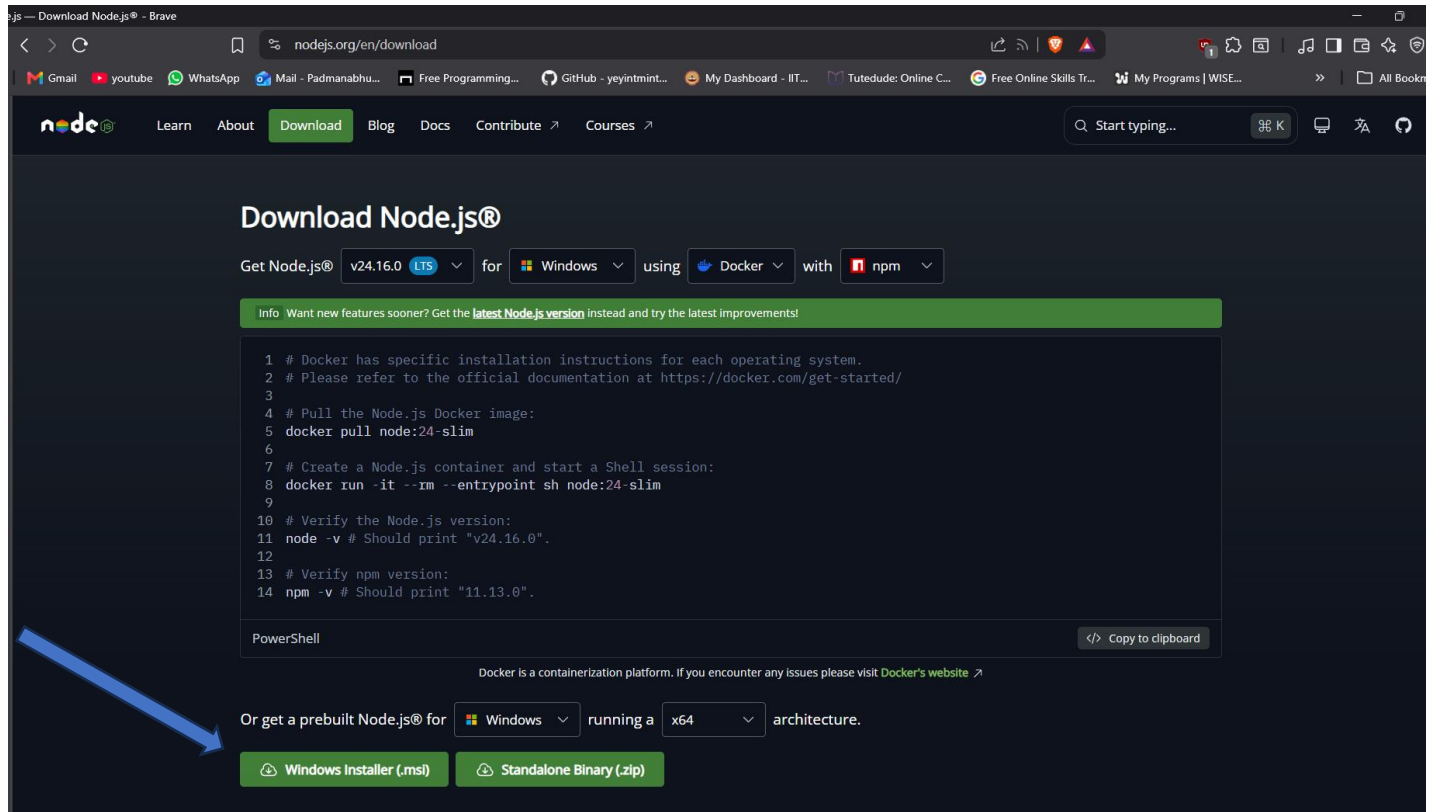
Explanation of the tags:

- body : centers form with gradient background
- .card : used to create form container
- form : used to collect user data
- input : used to take name email and password
- button : used to submit form
- onsubmit : triggers validation before submission
- function : defines form validation process
- getElementById() : used to access form fields
- .value : gets input values email
- Pattern : checks valid email format
- strong pattern : checks password strength
- alert() : shows success message when all validations pass
- return false : prevents submission if invalid
- return true : allows submission if valid
- innerText : displays error messages

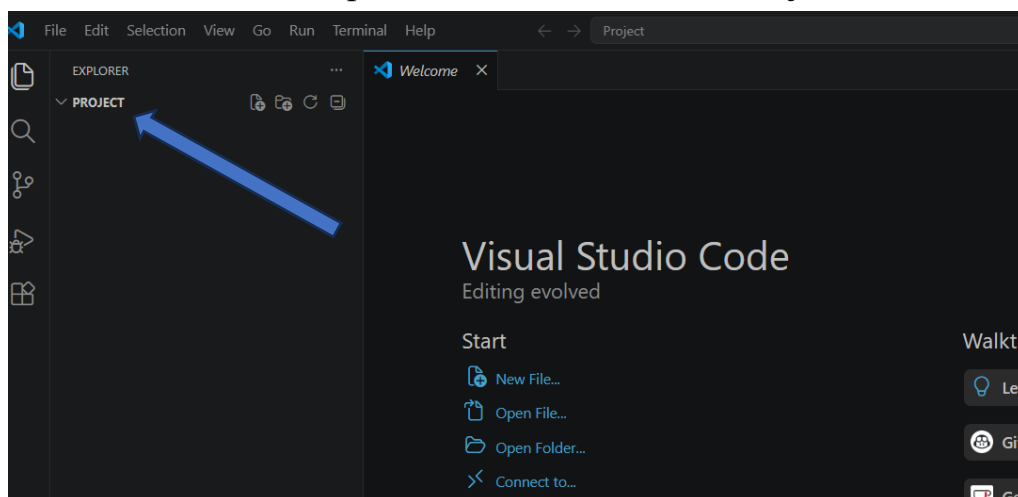
WEEK-12

12.Aim:Steps for installing electron in a Project

Step-1:Browse to [Node.js](https://nodejs.org/en/download) download website.Now Download the LTS version good stability and install it.



Step-2:To install **electron** in a project.Now create a folder with the project name and open it in VSCODE.For example,Here I used the name Project for the folder and opened VSCODE.



Step-3:Now open the terminal and start typing the following commands

1.npm init -y

```
PS C:\Users\asus\Downloads\Project> npm init -y
"version": "1.0.0",
"description": "",
"main": "index.js",
"scripts": {
  "test": "echo \"Error: no test specified\" && exit 1"
},
"keywords": [],
"author": "",
"license": "ISC",
"type": "commonjs"
}
```

2.npm install electron --save-dev

```
npm help
• PS C:\Users\asus\Downloads\Project> npm install electron --save-dev

added 23 packages, and audited 24 packages in 5s

2 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
❖ PS C:\Users\asus\Downloads\Project>
```

Then two files will added to the folder.

```
> node_modules
{} package-lock.json
{} package.json
```

Step-4: For Example Lets do a mini Calculator.

Now add the codes into the following files

index.html

```
main.js index.html x render.js package.json
index.html > html > body > h1
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Mini Calculator</title>
5 <style>
6   body {
7     font-family: Arial;
8     text-align: center;
9     margin-top: 50px;
10    background: #f4f4f4;
11  }
12  input {
13    padding: 10px;
14    margin: 10px;
15    width: 150px;
16  }
17  button {
18    padding: 10px;
19    margin: 5px;
20    width: 60px;
21    cursor: pointer;
22  }
23  h2 {
24    margin-top: 20px;
25  }
26 </style>
27 </head>
28 <body>
29 <h1>Mini Calculator</h1>
30 <input type="number" id="num1" placeholder="Number 1">
31 <input type="number" id="num2" placeholder="Number 2">
32 <br>
33 <button onclick="add()">+</button>
34 <button onclick="sub()">-</button>
35 <button onclick="mul()">*</button>
36 <button onclick="div()">/</button>
37 <h2 id="result">Result: </h2>
38
```

main.js

```
JS main.js > ...
1 const { app, BrowserWindow } = require('electron');
2 function createWindow() {
3   const win = new BrowserWindow({
4     width: 300,
5     height: 400,
6   });
7   win.loadFile('index.html');
8 }
9 app.whenReady().then(createWindow);
```

renderer.js

```
JS renderer.js >  getValues
1  function getValues() {
2      let a = Number(document.getElementById("num1").value);
3      let b = Number(document.getElementById("num2").value);
4      return [a, b];
5  }
6
7  function add() {
8      let [a, b] = getValues();
9      document.getElementById("result").innerText = "Result: " + (a + b);
10 }
11
12 function sub() {
13     let [a, b] = getValues();
14     document.getElementById("result").innerText = "Result: " + (a - b);
15 }
16 function mul() {
17     let [a, b] = getValues();
18     document.getElementById("result").innerText = "Result: " + (a * b);
19 }
20
21 function div() {
22     let [a, b] = getValues();
23     if (b === 0) {
24         document.getElementById("result").innerText = "Cannot divide by 0";
25     } else {
26         document.getElementById("result").innerText = "Result: " + (a / b);
27     }
28 }
```

Step-5: Now the scripts attributes in package.json file to “start”: “electron .”

```
{ package.json > ...
1  {
2      "name": "newproject",
3      "version": "1.0.0",
4      "description": "",
5      "main": "main.js",
6      "scripts": {
7          "start": "electron ."
8      },
9      "keywords": [],
10     "author": "",
11     "license": "ISC",
12     "type": "commonjs",
13     "devDependencies": {
14         "electron": "^41.3.0"
15     }
16 }
17
```


Step-6: Now use the command `npm start` to run the project in electron environment.

Output:

